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Anabolic Actions of Parathyroid Hormone on Bone*

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I. Introduction

PHYSICIANS have traditionally considered PTH to be an agent that is catabolic to the skeleton. However, as early as 60 yr ago, Albright and his colleagues (1) and later Selye (2) noted that the peptide extract could also be anabolic to the skeleton, producing increases in bone tissue in animals. The potential for an agent that can increase bone mass and hence reverse the skeletal defect in patients with osteoporosis is great, particularly if in doing so it also repairs microarchitectural damage. The agents currently available in the United States for treatment of osteoporosis, estrogens and salmon calcitonin, primarily stabilize bone mass and prevent further loss of bone, although a transient small increment in mass is often reported, particularly in patients with elevated levels of bone remodeling. This increase is not a true anabolic effect but is related to the temporal effects on turnover in which resorption declines initially followed by a reduction in formation that may take several months. Fluoride, in contrast, does increase bone mass by a true anabolic action, but there is controversy about the quality of the bone formed. Two recent controlled studies failed to find a reduction in fracture recurrence (3, 4), although the dose may have been excessive, and consequently, fluoride is currently not approved for treatment of osteoporosis in the United States. Thus, if PTH is capable of improving bone mass and quality, the peptide will have a potential therapeutic role in osteoporosis. This review summarizes the data available to date, starting at the cellular level.

II. Target Cells for PTH in Bone

Despite the fact that the major action of PTH in bone is to stimulate osteoclastic bone resorption, a compelling body of evidence has been published over the last two decades indicating that cells of the osteoblast lineage are the principal, but not necessarily the only target (5, 6). A detailed morphological description of a distinct "PTH target cell" (PT cell) was provided recently by Rouleau *et al.* (7, 8). Located among clusters of differentiated osteoblasts, this is a mononucleated cell with a large soma and long cytoplasmic processes extending toward the bone surface and also toward nearby blood vessels. Based on a positive reaction for alkaline phosphatase and the occasional observation of hybrid cells that exhibited ultrastructural features of both the PT cell and differentiated osteoblasts, Rouleau *et al.* (7, 8) concluded that the PTH-responsive cell is indeed of the osteoblast lineage.

The concept that osteogenic cells, rather than the osteoclast, were the primary targets for PTH as well as for other agents that stimulate bone resorption was first formally proposed in a classic paper by Rodan and Martin in 1981 (9). Experimental evidence in support of this hypothesis was obtained when investigators developed successful techniques for studying functionally viable osteoclasts *in vitro*. Thus, Chambers and his colleagues (10) showed that when osteoclasts were physically separated from other bone-derived cells on a devitalized slice of bone, they did not respond to PTH. However, when primary cultures of osteoblasts or osteoblast-like cell lines were added to the culture, PTH then

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induced an increase in resorption (11). These functional studies were supported by autoradiographic evidence suggesting a lack of specific PTH binding in osteoclasts (6, 7). Subsequent studies from Chambers' group and others (11–14) suggested the existence of a soluble osteoclast-stimulating factor(s) released by osteoblasts in response to PTH. To date, the identity of this factor has remained elusive, although recent reports have indicated that rather than being soluble, it may be membrane- or matrix-bound (15).

This body of work shifted the emphasis away from the osteoclast as a target cell for PTH despite earlier immunocytochemical evidence for this in fixed bone sections (16). However, in the past 2 yr, evidence has been presented by three independent groups that there are also PTH receptors in osteoclasts. Duong *et al.* (17) reported that bovine PTH(1–34) had an *inhibitory* effect on bone resorption by purified cultures of chick osteoclasts. A stimulatory response was observed, however, in less pure populations with a greater density of contaminant osteoblasts. Moreover, these authors presented autoradiographic evidence for the existence of PTH receptors in both avian and mammalian osteoclasts, postulating that these receptors mediate the observed direct inhibitory effects. Teti *et al.* (18) also described specific binding of radioiodinated PTH to avian osteoclasts, and using a biotinylated PTH binding method amplified by fluorescein isothiocyanate-labeled avidin, Agarwala and Gay (19) demonstrated binding of bovine PTH(1–84) to living chick osteoclasts. In the latter case, several characteristic properties of receptor-mediated hormone binding were described, including saturability, time and temperature dependence, and hormone specificity.

We must, therefore, at present acknowledge the intriguing prospect that both osteoblasts and osteoclasts express functional PTH receptors. This raises the possibility of dual regulatory mechanisms for PTH action on bone with the peptide perhaps exerting a direct inhibitory effect on the osteoclast, and an indirect stimulatory effect mediated by osteoblasts.

III. *In Vitro* Studies on the Anabolic Effects of PTH

A. Organ and cell cultures

While the earliest organ culture studies quickly confirmed PTH's potent bone resorbing activity (20), demonstration of its anabolic activity by this technique has not been so easy. Using Gaillard's original method of culturing embryonic mouse radii, Herrmann-Erlee *et al.* (21) showed that synthetic bovine PTH(1–34) ($\leq 5 \times 10^{-9}$ M) increased osteoblast number and osteoid formation in more than half of the samples studied. However, PTH decreased collagen synthesis in cultured fetal rat calvariae (22), an observation that has been confirmed in neonatal mouse calvariae (23) and, more recently, in fetal rat parietal bones (24), and is attributed to a PTH-induced inhibition of steady-state levels of procollagen messenger RNA (mRNA) (25, 26). This inhibitory effect of PTH on differentiated osteoblasts was corroborated by reductions in alkaline phosphatase activity (27, 28), osteocalcin synthesis, and collagen synthesis in cultured osteoblastic cells (26, 29–31). However, although PTH decreases osteoblastic synthesis and secretion of collagen and other peptides, it appears to stimulate osteoblast replication as measured by the incorporation of [^{3}H]thymidine into DNA in bone organ cultures (32, 33). This observation suggests that the anabolic effect of PTH might be mediated by an increase in the proliferation of osteoblast precursors or osteoprogenitor cells (34), which has been confirmed in some, although not all (35), cell culture models. Low concentrations of PTH stimulate proliferation in rat osteosarcoma-derived osteoblast-like cells (ROS 17/2, UMR 106), rat calvarial osteoblasts (28, 36), and in primary cultures of human bone cells of osteoblast phenotype, and human osteosarcoma cells (34). Interestingly, there is evidence to suggest that the PTH-induced proliferation of osteoblast precursors may be mediated by the osteoblasts themselves in a process that requires cell-to-cell contact (37). In general, the effects of PTH on mitogenesis are associated with a reduction in the expression of the osteoblast phenotype. However, an anabolic effect of PTH on bone that is independent of its mitogenic action should not be discounted. In this regard, Howard *et al.* (38) and, more recently, Canalis and his colleagues (33) have demonstrated that, as is the case *in vivo* (see below), the manner in which PTH is administered *in vitro* profoundly influences its effects on bone formation. Thus, in cultured fetal rat calvariae, continuous treatment with PTH for 24–72 h inhibited type I collagen production as previously reported (22, 26), but transient exposure for 24 h followed by incubation in PTH-free medium resulted in a stimulation of collagen synthesis (33). Furthermore, although [^{3}H]thymidine incorporation was increased in both circumstances, neither the inhibitory nor the stimulatory effects of PTH on collagen synthesis appeared to be dependent on mitogenesis because neither was affected by the presence of the inhibitor of DNA synthesis, hydroxyurea. The stimulatory effect on collagen synthesis was also observed in bones that had been stripped of periosteum, which contains the principal source of osteoblast precursors and is the site of most of the mitogenic effects of PTH. These results imply that PTH can have stimulatory effects on differentiated osteoblasts as well as on the differentiation of preosteoblasts. This is supported by the fact that, contrary to the studies mentioned earlier (27, 28), some osteoblastic cells respond to PTH with an increase in alkaline phosphatase, a time-honored marker of the differentiated osteoblast (39, 40). As a general note, it seems that the response of different osteoblast cultures to PTH is variable and depends crucially on whether they are primary cultures or cell lines, the species and age of the source, and the initial degree of differentiation in culture (41).

B. Role of growth factors in orchestrating the anabolic action of PTH

Recent studies in several laboratories have provided convincing evidence that the PTH-induced stimulation of bone formation is mediated by local growth factors, with insulin-like growth factors (IGF)-I and -II and transforming growth factor-β (TGFβ) being the primary candidates. IGF-I and II

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enhance collagen production by two independent methods: stimulation of osteoblast replication and increasing the synthesis of type I procollagen mRNA by differentiated osteoblasts (42). PTH stimulates the release of IGF-I from fetal rat calvariae (33) and of both IGF-I and II from mouse calvariae (43) and also increases the transcript and polypeptide levels of IGF-I in osteoblast-enriched primary cell cultures (44). Moreover, the stimulatory effect of transient PTH administration on type I collagen synthesis in fetal rat calvariae is abolished by IGF-I neutralizing antibodies (33), an effect that is again independent of PTH's mitogenic action, which was unaffected by the antibodies. In addition to increasing osteoblastic synthesis of IGFs and of IGF binding proteins, PTH may also increase the sensitivity of osteoblasts to IGF-I by increasing the affinity and/or the number of IGF receptors. This provides a mechanism for the synergistic effects of PTH and IGF-I on bone formation observed both *in vitro* and *in vivo* (45).

Members of the TGFβ supergene family, which include TGFβ 1 and 2 and bone morphogenetic proteins 2–7 are clearly important bone growth factors (46–48). Indeed, bone is one of the most abundant sources of TGFβ in the body (49). TGFβ is synthesized and secreted by osteoblasts in a biologically inactive, latent form and is incorporated in the extracellular matrix. Osteoblasts not only produce TGFβ but they also possess high affinity receptors for it, providing the opportunity for autocrine stimulation of osteoblast replication (50). Moreover, a recent study suggests that TGFβ can increase the number of functional PTH receptors in osteoblasts (51). TGFβ is a potent stimulator of bone formation both *in vitro* (52, 53) and *in vivo* (54–56). While PTH does not appear to regulate osteoblastic synthesis of TGFβ (44), its initial action to stimulate osteoclastic bone resorption results in the liberation of significant amounts of TGFβ from the matrix. Furthermore, the acidic microenvironment in the subosteoclastic resorption zone appears to be capable of converting TGFβ from latent to active form (57). Once released from bone, the activated TGFβ could presumably stimulate osteoblast chemotaxis, replication, and differentiation in the vicinity of the resorption lacuna and possibly also act to inhibit osteoclast activity and recruitment (46, 58). TGFβ is therefore an ideal candidate for the putative "coupling factor" that tethers bone formation to bone resorption in the remodeling cycle. Other osteoclast-accessible, matrix-bound growth factors that could act in a similar fashion to mediate an anabolic effect of PTH include IGF-I and II (43, 59), platelet derived growth factor (60), and acidic and basic fibroblast growth factors (61, 62), although evidence for involvement in this PTH effect is currently only available for the IGFs. It is worth noting, however, that while this indirect, osteoclast-dependent mechanism of stimulating osteoblastic bone formation may well be operative, it is not essential for eliciting PTH's anabolic action. Tam *et al.* (63) showed that in short-term experiments, intermittent injection of PTH in rats resulted in an increase in bone-forming surfaces and apposition rate without a change in resorption surface and, more recently, Hock *et al.* (64) reported that the anabolic action of PTH in rats was unaffected by treatment with the resorption inhibitors, calcitonin and dichloromethylene bisphosphonate (see below).

C. Second messengers

The fact that PTH can have both catabolic and anabolic effects on the skeleton raises an intriguing question: How does a single hormone achieve such separate and opposing actions in bone, especially as both actions appear to be mediated either directly (formation) or indirectly (resorption) by the same cell type, namely the osteoblast? As mentioned earlier and discussed in more detail below, part of the answer may be the differential response of the target cells depending on whether PTH is delivered continuously, which appears to favor resorption, or in an intermittent fashion which appears to favor formation. An alternative explanation may lie in the ability of PTH to activate at least two different intracellular signal transduction pathways in osteoblasts (64a) (Fig. 1). These are cAMP-dependent protein kinase A (cAMP/PKA) (65, 66), and the calcium/protein kinase C (Ca^2/PKC) pathways (67, 68). At the present time it is not clear which cellular functions in osteoblasts are regulated by which pathway (69). Studies in this area are complicated by the fact that both osteoblasts and osteoclasts "use" the same transduction mechanisms. Thus, stimulation of the cAMP pathway in osteoblasts by experimental agonists, such as forskolin, could theoretically result in stimulation of resorption but this effect may be counterbalanced by simultaneous activation of the cAMP pathway in osteoclasts, which is inhibitory in effect (70, 71). The same problem applies to the Ca^2/PKC pathway, which has recently been shown to be present in osteoclasts (72–74), as well as osteoblasts (67, 68). Perhaps reflecting this difficulty, in two recent publications it has been concluded that the cAMP pathway in osteoblasts is either of *major* (75) or *minor* (76) importance in conveying the stimulatory effect of PTH on bone resorption. Similarly, there is little definitive information on the second messengers responsible for mediating the anabolic action of PTH. Studies with osteoblast-like cell lines indicate that both the cAMP/

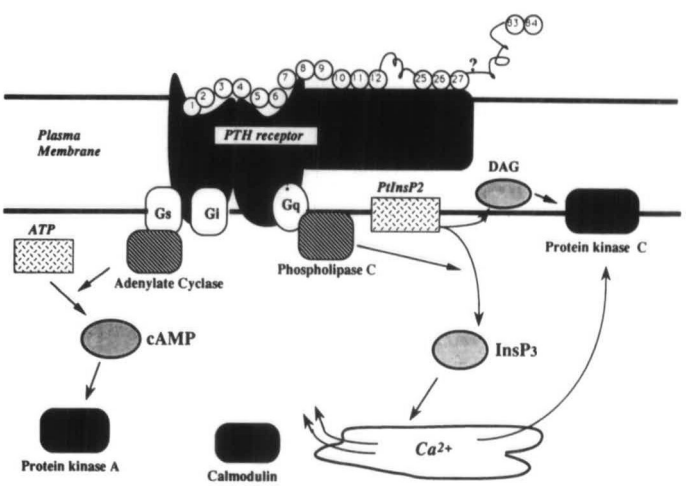

FIG. 1. A model for PTH signal transduction, incorporating both the cAMP/PKA and Ca^{2+}/PKC pathways. [Reproduced with permission from A. Fujimori *et al.*: *Endocrinology* 130:29–36, 1992 (64a). ©The Endocrine Society.]

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PKA and the Ca^{2+}/PKC systems play a role in the regulation of proliferation as well as phenotypic expression (75, 77, 78). Based on experiments in which the cAMP and the intracellular Ca^{2+} pathways were selectively activated by different PTH fragments, Lowik *et al.* (79) suggested more than 5 yr ago that osteoblasts contain two distinct receptors with one being coupled to adenylate cyclase and the other involved in the elevation of intracellular Ca^{2+}. However, this hypothesis remains unsubstantiated (69). The precedence for subclasses of receptors conferring specificity for the second messenger pathway exists in other systems. For example, the α-adrenergic receptor is classified into α_1- and α_2-subtypes with the former assigned to the phosphatidylinositol pathway and the latter to the inhibition of adenylate cyclase (80, 81). Modeling of the PTH receptor site in canine kidney (82) suggests the existence of either two separate receptors for PTH(1–84) and PTH(1–34) with similar affinities, or alternatively, a second binding site for PTH(1–84) on the same receptor. Another possibility is that the local conditions in and around osteoblasts could influence the balance between the two pathways. For instance, in rat osteosarcoma cells (ROS 17/2.8) it is known that the presence of retinoic acid influences inositol polyphosphate metabolism, that a decrease in pH activates phospholipase C, and that PTH-stimulated adenylate cyclase activity is enhanced by glucocorticoids (81).

Although it is now firmly established that growth factors play a role in mediating the anabolic effects of PTH, there may be other underlying mechanisms. Finkelman *et al.* (83) recently reported that PTH enhances the proliferation of TE-85 human osteosarcoma cells by a mechanism that does not involve either increased cAMP production or enhanced secretion of IGF-I, IGF-II, or TGFβ. The calcium channel blocker, verapamil, reduced cell growth, both basal and PTH-stimulated, but a significant stimulation was still observed, suggesting that, while a rise in intracellular calcium may be involved in mediating the PTH response in this model, it is not the only second messenger.

While studies on the anabolic action have necessarily focused on osteoblasts and bone formation, we should not overlook the potential that "anticatabolic" effects might also alter the balance between formation and resorption in favor of the former. As mentioned above, Duong *et al.* (17) have reported inhibitory effects on the activity of osteoclasts in osteoblast-deficient cultures. Using the resorption pit assay, we have found evidence for an inhibitory effect on osteoclast function even in the presence of osteoblasts (84). While osteoclasts were activated by PTH and the average number of pits per resorption focus was increased, the median plan area of the individual resorption pits was substantially reduced. Such an effect is consistent with histomorphometric data in patients with primary hyperparathyroidism in which there is increased activation frequency of bone remodeling units but a reduction in the depth of the resorption lacunae (see below). Incidentally, Fenton *et al.* (85) have recently reported that a highly conserved pentapeptide region of PTH-related protein (PTHrp[107–111]) is a potent inhibitor of osteoclastic activity *in vitro*.

In conclusion of this section, these basic studies have elucidated several possible mechanisms for a direct anabolic action of PTH on the skeleton. These include cytokine-mediated stimulation of osteoblastic proliferation and activity and, possibly, inhibition of osteoclast activity (Fig. 2). The remainder of this review will focus on PTH action in intact animals and humans.

IV. Animal Data

The majority of data supporting an anabolic effect of PTH comes from animal studies. Although the rat has been the principal species in which the effects of PTH have been evaluated using a variety of models of bone loss, data are also available for the dog. There is no report of an anabolic action of PTH in any primate species as yet.

A. Early discoveries

Early results were all performed using parathyroid extract (PTE) rather than purified PTH. The earliest report of increased bone density by PTE (8 U sc daily) in rats was that of Bauer and associates (1) but they did not comment on the significance of the finding. Convincing evidence that PTE could stimulate bone formation, in addition to bone resorption, was obtained histologically shortly thereafter by Selye (2) and Pugsley and Selye (86). They observed increased proliferation of bone cells, both osteoclasts and osteoblasts, shortly after a single injection of 20 U PTE to rats. If the treatment was continued, an increase in bone tissue resulted

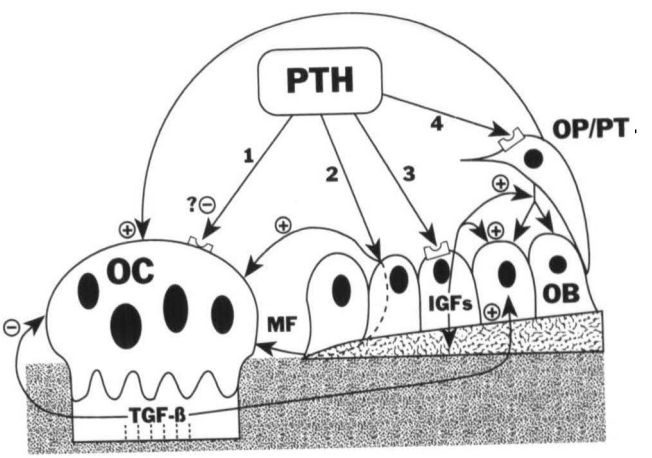

FIG. 2. Diagram illustrating some of the diverse effects of PTH on bone. Stimulation of osteoclastic bone resorption is mediated indirectly via osteoblasts or the putative parathyroid (PT) cell (*pathways 2 and 4*). The signal from osteoblasts/PT cells to osteoclasts may be conveyed by soluble osteoclast-stimulating factor(s) or possibly a membrane or matrix-bound factor (MF). In addition to this indirect communication, there is growing evidence that PTH may act directly on osteoclasts, perhaps exerting an inhibitory influence (*pathway 1*). The anabolic action of intermittent PTH administration is likely mediated by growth factors, including TGFβ, released from the matrix and activated by osteoclasts, and by IGF's I and II, produced by the osteoblasts themselves (*pathway 3*). The growth factors enhance both the replication and activity of osteoblasts. See text for further details. OC, Osteoclast; OB, osteoblast; OP/PT, osteoprogenitor/"parathyroid cell"; MF, membrane- or matrix-bound factor.

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and the osteoclast number declined to normal. Interestingly, they noted that treatment with lower doses of PTE (5 U on alternate days) led to proliferation of osteoblasts only. Selye (2) suggested that PTH given in small doses, would lead to a stimulation of osteoblasts and bone apposition without a previous osteoclast formation and fibrous transformation of the bone marrow. Using a similar dose of PTE in rats, Seyle's results on the stimulation of osteoblasts by PTH were confirmed by Jaffe (87) and Burrows (88). Shelling *et al.* (89) directly measured rat femurs and demonstrated an increased ratio of dry femur weight in rats treated with PTE daily for 14 or 21 days. Using a single dose of 1000 U PTE, Heller *et al.* (90) showed a rebound of osteoblast number and activity in 4–7 days after initial inhibition, suggesting a long lasting effect of PTE. Biochemical analysis of matrix materials and radiological density measurements were later added, confirming the anabolic action of PTH in the 1970's (91, 92). Increments in the calcium accretion rate into the skeleton of the order of 45% were noted after chronic administration of PTE in rats (93). A positive calcium balance with daily PTE administration was observed by Cramer *et al.* (94). Using rats on normal and low phosphate diets, the positive calcium balance was observed in 5-day periods after injection of PTE (50 U twice daily for the first 3 days). The early results on the anabolic effect of PTH on animal skeletons have been extensively reviewed previously by Parsons (95).

B. Quantitative determination of the anabolic action of PTH

During the last decade improved analytical techniques and the wide availability of purified PTH fragments led to a renewed interest in the study of the anabolic action of PTH on bone based on sensitive quantitative determinations of the skeleton and its remodeling processes. The apparent success of daily administration of low doses of PTH to humans (96–98) (see below) and in controlled data using dogs (99, 100) further prompted this interest.

In rats, Hefti *et al.* (101) showed an increase in total body calcium, as well as calcium content in 11 different bones, by administering 75 U human PTH (hPTH) (1–34)/day. Tam *et al.* (63) reported an increase in bone formation surface and trabecular bone volume, but not in resorption surface, in rats treated intermittently with 0.4–102.4 U/rat/day of PTH molecules for 12 days. Gunness-Hey and Hock (102), using a novel method of separating cortical and trabecular bone, demonstrated a significant dose-dependent increase in cancellous bone, but not in cortical bone, after PTH injection. This result has been confirmed by several other authors (103–107). The use of histomorphometric methods to quantitate the gain in cancellous bone volume and the changes in bone formation surface or mineral apposition rate have contributed a greater understanding of the mechanism of PTH action (63, 105, 108–114). Results from these experiments showed that the anabolic action of PTH, when given by daily injection, is consistent and proportional to the dose used in the experiment. Injection of PTH for as short a treatment period as 12 days (115) or for as long as 175 days (103) has been shown to effectively increase cancellous bone mass. However, differences in the rapidity and duration of the responses are not known because all information was gathered at the end of the experiments (*i.e.* 12–175 days). Both male and female rats respond to PTH treatment equally well (104, 113) and the age of the rats that respond to PTH can vary from prenatal (92) to 2 yr (113, 116). An interruption in the daily therapy of 4 days results in significant loss of the stimulated bone formation activity, and a consequent loss of bone mass can be observed as early as 8–12 days after the withdrawal of the hormone (111). In all published data the anabolic action of PTH is most evident in cancellous bone while increases in cortical bone mass are either not observed or less pronounced. While the bulk of the data was collected from cancellous enriched regions of tibia or femur, a few studies have observed similar changes in vertebrae (112, 113, 117). The percentage improvement in the mass of various bones, and even in different parts of the same bone, may be related to the different proportions of cortical and trabecular bone and to the metabolic activity in each type of bone. The notion, interpreted from human studies, that cancellous bone is increased by PTH at the expense of cortical bone (96, 98) has not been seen in the rat (63, 118). However, this may be a limitation of the model, since in the cortical bone of rats there is little Haversian remodeling, which may be a prerequisite for such an effect. The dietary calcium intake is also proportionally much higher in the laboratory rat than in humans, making it unnecessary to remove calcium from cortical bone to support cancellous bone formation elsewhere. Unlike the situation suggested for sodium fluoride, the bone formed under stimulation by PTH appears to be of normal quality. The increases in mass are accompanied by an improvement in biomechanical strength of vertebral bodies and this remains even after normalization for both cross-sectional area and bone mass (114, 119). The studies involving treatment with PTH in rats are summarized in Table 1.

C. Preventive and curative treatment of osteopenia with PTH

Early experiments with PTH were performed in intact animals with normal skeletal status. Despite earlier concerns regarding their suitability, several rat models of osteoporosis have been developed, indicating that rats may indeed be useful in the study of the pathogenesis and treatment of osteoporosis in humans. For example, in rats, ovariectomy, orchidectomy, and immobilization of limbs can induce bone loss in the skeleton with increments in bone turnover as is seen in humans. The therapeutic agents used in humans, estrogen, calcitonin, and bisphosphonates (120, 121), reduce bone turnover and prevent further bone loss in the rat models as they do in humans without substantially restoring bone mass or architecture. When tested in various rat models of osteoporosis, intermittent injection of PTH both prevents loss and restores bone mass caused by ovariectomy (103, 104, 108–110, 113, 118), orchidectomy (104), and hemicordectomy (109). The magnitude of the increase (in percentage terms) in bone mass induced by PTH administration in osteoporotic animals is generally greater than that obtained in intact animals. Conventional treatment with antiresorptive agents will not restore lost bone mass or structure. Addition

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TABLE 1. Quantification of the anabolic action of PTH on rat bones

Animals	PTH	Dose (μg/kg/day)	Length of Treatment	Oper	Bone	Percentage change in			Ref.
						TBV	ash wt[e] or calcium	BMD	
6 mo ♀ Wis	bPTH 1-34	75[a]	3 wk	Int	Whole body		20		(110)
300 g ♂ SD	bPTH 1-84	0.4–102[a]	12 d	Int	Femur	0–60			(64)
4–6 wk ♂ SD	hPTH 1-34	20–100	12 d	Int	Femur		6–60		(103, 133) (107)
10 wk ♀ SD	hPTH 1-34	6.0	26 wk	Int	Femur		4		(104)
				Ovx			15		
5 wk ♂ SD	hPTH 1-34	80	12 d	Int	Femur		34 (Trab) 12 (Cort)		(105)
				Ochx	Femur		37 (Trab) 19 (Cort)		
5 wk ♀ SD	hPTH 1-34	80	12 d	Int	Femur		18 (Trab) 5 (Cort)		(105)
				Ovx	Femur		14 (Trab) 10 (Cort)		
4 wk ♂ SD	hPTH 1-34	80	12 d	Int	Femur	100	31 (Trab) 20 (Cort)		(106, 108) (112)
8 mo ♀ Wis	hPTH 1-34	60	35 d[b]	PTX +Ovx	Vert	70 (Trab) 30 (Cort)			(110)
				+Cx	Tibia	300 (Trab) 12 (Cort)			
3 mo ♀ SD	hPTH 1-34	80, 160	35 d	Ovx	Femur		34	13	(119)
					Vert	70	49	12	
					Tibia	550			
4 mo ♀ SD	hPTH 1-34	1.5	28 d[c]	Ovx	Tibia	46		4	(109, 111)
3 mo ♀ SD	hPTH 1-34	80	20 d[d]	Ovx	Femur		50	4	(114)
					Tibia	271 (Trab) 4 (Cort)			
					Vert		12	5	
70 d ♂ Wis	hPTH 1-34	4–120	30 d	Int	Vert	13–59	0–32		(115)
	hPTH 1-84	10–300	30 d	Int	Vert	11–45	5–24		
3 mo ♂	bPTH 1-34	80	21 d	Int	Vert			9	(114)
23 mo ♂			21 d	Int	Vert			14	(114)
24 mo ♀	hPTH 1-34	80	21 d	Int	Vert			16	(114)
24 mo ♀	hPTH 1-34	160	14 d	Ovx	Femur	178		44	(117)
5 mo ♀	rPTH 1-34	40	28 d	Ovx	Tibia	175	0	(109)	
					Femur	198	2		
						165	9		

Abbreviations: bPTH, Bovine PTH; wk, week; mo, month; d, day; TBV, trabecular bone volume; BMD, bone mineral density; Vert, lumbar vertebrae; Trab, trabecular bone; Cort, cortical bone; Int, intact; Ovx, ovariectomy; Ochx, orchidectomy; Cx, hemicordectomy; Oper, operation; SD, Sprague Dawley; Wis, Wistar.

[a] Unit of PTH/rat/day.

[b] Treatment begins 11 wk after the operation.

[c] Treatment begins 4 wk after the operation.

[d] Treatment begins 40 days after the operation.

[e] Ash wt. was measured as calcium content in either trabecular or cortical bone of the distal femur.

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of an anabolic agent to the antiresorptive treatment should in theory achieve both a reduction in bone resorption and an enhancement of bone formation. This hypothesis is supported by our recent study of the effects of treatment of bone loss in ovariectomized rats with a combination of estrogen and PTH. The results indicated that such combined treatment was not only capable of increasing cancellous bone volume but also of reestablishing trabecular connectivity (108) (Fig. 3). This ability to restore bone mass and structure previously lost (108, 109, 113) has significant implications for the treatment of established osteoporosis in humans, especially for the disease among postmenopausal women, the most common form of the disorder, estimated to affect some 20 million women in the United States.

D. Intermittent vs. continuous administration of PTH

The discrepancy of the findings between PTH injection and severe primary hyperparathyroidism where bone loss was evident (see below) remained a puzzle for a long time (95) until experiments conducted in both dogs and rats suggested convincingly that the difference may, at least in part, relate to differences in the modes of administration of PTH. In dogs, continuous low dose hPTH (1–34) given by infusion (0.05 U/kg/h), mimicking chronic (mild) primary hyperparathyroidism, was associated with increments in osteoid surface and bone resorption at the tissue level, without an increment in bone volume. The authors suggested that this result could be explained by an increase in coupled turnover with equal increments in bone formation and resorption (122). In a separate series of studies, daily sc injection of hPTH (1–34) (15 U/kg/day) was compared with continuous infusion (4.5 U/kg/day) in greyhounds (99, 100). Daily injection of PTH significantly increased indices of bone formation (osteoid surface, plasma alkaline phosphatase, and rate of skeletal accretion of calcium) and resorption (number of osteoclasts and resorption surfaces) and increased trabecular bone volume in the ilium. The daily doses of PTH used in this study were similar to those used in human studies, and the resulting changes in bone parameters were also similar to those in humans (96) and, thus, the findings may be relevant to the human. On the other hand, continuous

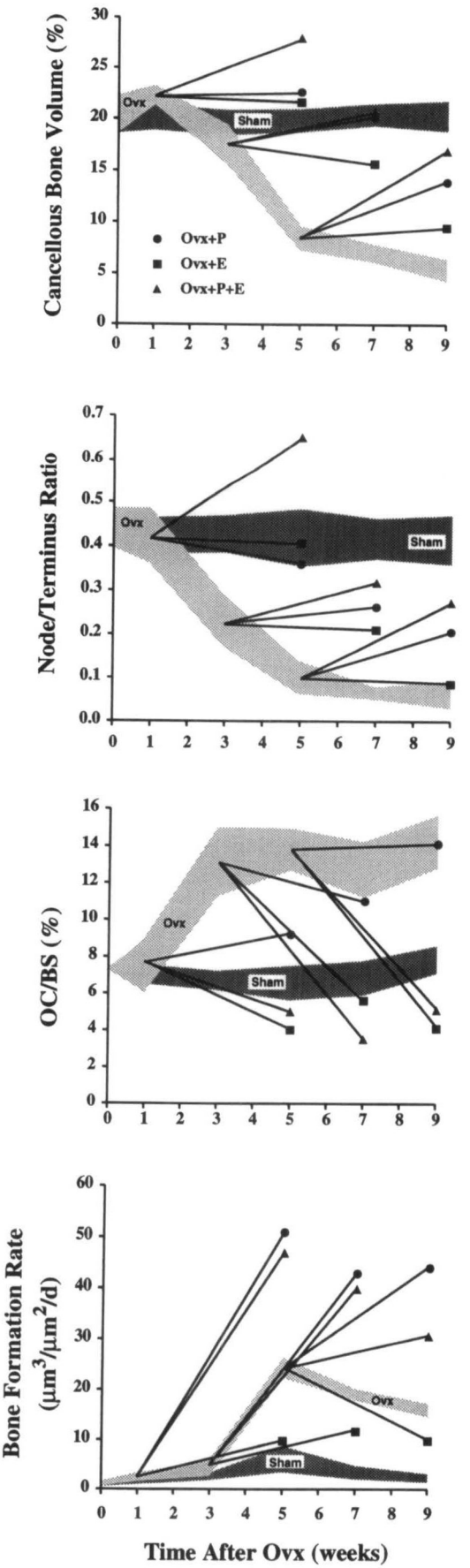

FIG. 3. Effects of treatment of bone loss in ovariectomized (Ovx) rats with a combination of PTH amd estrogen. Five-month-old rats were ovariectomized at time 0. Groups of rats were treated with 40 μg rat PTH(1–34)/kg/day of and/or 30 μg/kg/day of 17β-estradiol for 28 days starting at 1, 3, or 5 weeks after ovariectomy. Cancellous bone volume, node/terminus ratio (an index of trabecular connectivity), osteoclast surface, and bone formation rate were measured in the secondary spongiosa of the proximal tibia after *in vivo* tetracycline labeling. The *dark and light shaded* areas represent the results from the rats that underwent sham operation or ovariectomy, respectively, and were treated only with vehicles. The *circles* represent Ovx animals treated with PTH only, the *squares* represent Ovx animals treated with estradiol only, and the *triangles* represent Ovx animals treated with the combination of PTH and estradiol. Note that even when treatment was delayed for 5 weeks, the combined treatment not only significantly increased cancellous bone volume but also restored trabecular connectivity that had previously been lost. The efficacy of the combined treatment appears to be due to its ability to maintain a high bone formation rate (PTH effect) while decreasing the osteoclast surface (estrogen effect). [Drawn from data presented in Ref. 108.]

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infusion of PTH given by Alzet osmotic pumps, using similar doses of PTH, did not significantly increase trabecular bone volume or calcium accretion rate, and only osteoclast surfaces were clearly increased. Tam *et al.* (63) gave rats various doses of bovine PTH (1–84) by either daily injection or continuous infusion. In this experiment, bone formation surfaces were elevated by both methods of administration but resorption increased only if PTH was given by continuous infusion. The net result was an increase in cancellous bone volume when PTH was given by daily injection. Infusion, on the other hand, maintained cancellous bone volume at low doses but caused a decrease in cancellous bone volume when given in high doses. These observations were recently confirmed by Hock and Gera (123) who concluded that both modes of administration increased bone formation in the rat but only intermittent PTH (daily, sc) increased bone mass consistently (123). Thus, the destructive effects of hyperparathyroidism on bone, as reported in early studies of this disease, may be comparable with the high dose experiments while the more recent findings that trabecular bone mass is preserved in mild hyperparathyroidism may be the result of the lower endogenous dose of PTH. This will be discussed in greater detail below.

E. Comparison with skeletal effects of other anabolic agents

Many agents, such as fluoride, several vitamin D metabolites, prostaglandin E_2 (PGE_2), TGFβ, GH, and IGF-I, show the potential to improve bone mass. Although it is generally agreed that fluoride treatment increases cancellous bone volume in chicken, mice, rats, dogs, and pigs as well as in humans, it is difficult to use because of the relatively narrow therapeutic window (124), and the poor quality of bone raises the question, as yet unproven, whether fluoride can ever reduce fracture frequency. Studies using PGE_2 as an agent for new bone production have also yielded very consistent results. PGE_2 appears to activate remodeling and shift bone balance in favor of formation (125, 126) although some *de novo* formation of bone may also occur. This is somewhat different from the apparent effects of intermittent PTH in which uncoupled formation occurs without the need for prior resorption. There are no published data in which a direct comparison of the use of these two agents has been performed. The net effects of long term *in vivo* administration of vitamin D metabolites on bone metabolism can be quite varied. The differences appear to be modulated by the types and dose of vitamin D metabolites used, by dietary calcium intake, and possibly the osteoblast sensitivity, as suggested by Erben *et al.* (127). Studies using TGFβ, GH, and IGF-I as anabolic agents of bone tissue in animals have been scarce (45, 107, 116, 128–130). GH and IGF-I are likely to behave anabolically while TGFβ seems less likely to be an anabolic agent. While GH and IGF-I would be potential candidates for the treatment of osteoporosis, one has to be mindful that these agents have widespread actions on extraskeletal tissues and could be associated with significant undesirable side effects (131). We believe that a better and more efficient approach is to increase the local concentration of growth factors in bone by means of a bone-seeking, growth factor "stimulator," such as PTH.

F. Mechanism of action

Although we do not know the exact mechanism underlying the anabolic action of PTH, there are data that exclude some possibilities. The effects of PTH are not due to compensatory changes in calcitonin secretion because thyroparathyroidectomized rats respond to PTH challenge with an increase in bone mass (91, 109). Hypercalcemia *per se* has no stimulatory effect on bone formation because injection of calcium chloride, instead of PTE, elicits no bone formation response (132). A PTH-induced increase in PGE_2, an anabolic agent in its own right, is unlikely to be the mechanism because coadministration of indomethacin with PTH failed to abolish or modify the anabolic effects of PTH in rats (133). Prior resorption does not appear necessary, as noted above, since stimulation of bone formation without resorptive activity can be seen in rats when PTH is administered by daily injection (63). Furthermore, coadministration of a resorption inhibitor, calcitonin, dichloromethylene bisphosphonate, or estrogen, with PTH fails to block the anabolic response of bone to PTH (105, 108). In fact, as noted above, the combined effects of estrogen administered with PTH are significantly better than either alone, both in terms of bone mass and trabecular structure (Fig. 3 and Ref. 109). The anabolic effects of PTH are not linked to longitudinal growth because aged animals with closed growth plates can still respond to PTH (113) and many studies, with growing animals, showed no specific effect of PTH on long bone length (112, 113, 134).

One of the main functions of PTH is to stimulate the activity of the 1α-hydroxylase in the kidney resulting in an elevation of circulating levels of 1,25-dihydroxyvitamin D :ob1,25$(OH)_2$D] (135). It has been proposed that the increase in 1,25$(OH)_2$D levels may improve intestinal calcium absorption and thus, create a positive calcium balance. 1,25$(OH)_2$D is also known to directly affect the maturation of both osteoblastic and osteoclastic lineage cells (135, 136). It is perhaps not surprising, therefore, that the PTH-induced increment in circulating 1,25$(OH)_2$-D levels has been proposed as necessary for the anabolic action of PTH (113, 118). Indeed, 1,25$(OH)_2$D levels in PTH-treated animals are elevated (108, 113, 118, 137). In one experiment in which rats were given a vitamin D-depleted diet, PTH failed to induce improvement in the bone apposition rate (138) suggesting that increased 1,25$(OH)_2$D may be important for the expression of at least some of the anabolic action of PTH. However, other studies have suggested a less important role for 1,25$(OH)_2$D (106, 108). Supplementation of 1,25$(OH)_2$D to PTH-treated animals does not appear to enhance the anabolic action of PTH (107). Combined treatment of estrogen and PTH produced the greatest improvement in bone mass yet the 1,25$(OH)_2$D level (which is paradoxically reduced by estrogen in rats) (108) was not greater than control levels. This suggests that an absolute elevation of 1,25$(OH)_2$D levels may not be necessary for the increase in bone mass to occur (109).

According to histomorphometric observations, PTH in-

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creases bone formation by extending mineralization surfaces and/or the rate of mineral apposition. This is likely the result of increased osteoblast precursor proliferation and/or maturation. As discussed above, there is some indirect *in vitro* evidence as to how this may be achieved. More details of these studies were presented in the last section. This line of reasoning is highly attractive because it has been reported that the anabolic effect of hPTH depends on the presence of GH, a significant positive modulator of IGF-I secretion (107). Further experimentation will be needed to decipher the interaction between PTH, IGF-I, IGF-II, and GH in the anabolic action of PTH in rats.

In concluding this section, intermittent injection of PTH in animals, especially in rats, exhibits a consistent anabolic action on bone tissue. These results suggest that an animal model can provide relevant information on the development of effective PTH treatment regimens for osteoporosis in humans, particularly as PTH administration can apparently restore previously lost trabecular bone mass and structure.

V. Lessons from Primary Hyperparathyroidism: Effects of Endogenous PTH on Bone in Humans

Secondary hyperparathyroidism is known to occur as a homeostatic mechanism in response to decreasing calcium absorption efficiency and decreased 1α hydroxylation of 25-hydroxyvitamin D (138a). This phenomenon has been implicated in age related-bone loss. In addition, primary hyperparathyroidism is often cited as a risk factor for osteoporosis, although in the most recent Consensus Development Conference, this was restricted to severe primary hyperparathyroidism (138b). In fact, recent studies of patients with mild primary hyperparathyroidism suggest that mildly elevated circulating levels of PTH may have beneficial effects, at least on cancellous bone. This section will review these recent findings.

When primary hyperparathyroidism was described 100 yr ago, it was recognized as a syndrome of bone pain, fractures, and renal stones (139–143). The radiologically evident destructive bone lesions and histological observations of severe unbalanced osteoclastic resorption (144–148) led to the belief that high levels of endogenous PTH exerted a catabolic effect on the skeleton. Many years later, increased awareness of this disease (149–152) and the availability of multichannel chemistry screening dramatically changed the mode of presentation of primary hyperparathyroidism. Hence, for the past 15–20 yr, presumably through this phenomenon of early recognition, a more protean milder clinical phenotype is seen most often. Patients are usually asymptomatic and discovered incidentally. Hypercalcemia is mild and clinically evident bone disease is quite uncommon (149, 153–157).

Several studies had suggested that a catabolic effect of PTH on the skeleton might lead to widespread demineralization and fractures even in the presence of mild disease (158–161). In fact, loss of cortical bone has clearly been documented in both severe and mild primary hyperparathyroidism; there is decreased bone mineral density at appendicular sites rich in cortical bone (157, 161–164) and cortices are thinner in iliac bone biopsies (163, 165–167). In contrast, the effect of excess endogenous PTH on cancellous bone is more controversial. While cancellous bone loss at the spine (162, 168) and at the wrist (161) is suggested by some densitometric studies, recent findings indicate conservation of cancellous bone in this disease (157). Similarly, the suggestion that there is an increased vertebral fracture rate in primary hyperparathyroidism (158–161) has not been supported (164, 169, 170). In a recent study by Melton and associates (171), the overall fracture risk was increased in primary hyperparathyroidism compared to age- and sex-matched controls before, but not after, diagnosis. The discrepancies in the data on fracture could be explained, at least partially, by differences in patient populations, nutritional status, disease severity, study designs, and methodology (169). Clearly, the influence of mild primary hyperparathyroidism on fracture risk is an area where further research is needed.

The technique of bone histomorphometry has allowed evaluation of the skeletal effects of the modest elevations of PTH seen in the modern presentation of the disease. While the effects of the disease on bone mass and fracture incidence are still hotly debated, histomorphometric studies performed on different populations over the past 15 yr have failed to show evidence of cancellous bone loss in primary hyperparathyroidism (163, 166, 167, 172–177). Several mechanisms have been proposed to explain the conservation of cancellous bone volume. These include a coupled and balanced increase in bone turnover (176, 178) with increased osteoblastic activity and a prolonged formation period (176), and increased remodeling activation frequency with decreased erosion depth (172, 174). Static variables of bone turnover, such as osteoid surface and eroded surface, are increased in primary hyperparathyroidism suggesting increased bone turnover (173, 175–182). This has been confirmed by dynamic histomorphometric studies performed after skeletal labeling with tetracycline (167, 172, 173, 177, 183, 184). The increased turnover occurs primarily by means of increased activation frequency of new bone remodeling units (174, 183). Moreover, consistent correlations between static and dynamic indices of bone turnover and the mid-molecule of PTH, between these indices and urinary cAMP, and between eroded surface and serum calcium provide strong evidence for a causative role of PTH in the enhanced bone turnover (167).

An anabolic effect of PTH on bone has been suggested by animal studies as well as by clinical therapeutic trials. The observation of osteosclerosis in the metaphyses, skull, vertebrae, pelvis and femora of adults (185–191), and in the metaphyses of children and adolescents uncommonly affected with the disease (192–197) lend credence to this concept. Our recent experience with iliac bone biopsies in primary hyperparathyroidism has provided further evidence that PTH in moderate excess has a protective effect on cancellous bone (157, 167, 197a). The technical developments that have occurred in the field of bone histomorphometry in recent years allow the study of not only mass and turnover, but of bone microarchitecture (198–202). We have

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applied this approach to primary hyperparathyroidism to obtain more detailed information on the changes in bone structure. In a study of 27 hyperparathyroid subjects, we observed a striking contrast in the microarchitecture of cortical and cancellous bone between normal individuals and those with mild elevations of endogenous PTH (157, 167, 197a). Cortical thickness was reduced in the hyperparathyroid subjects, confirming previous observations (163, 165) but cancellous bone volume, trabecular number, and importantly trabecular connectivity, as assessed by strut analysis, were higher than in control subjects (Fig. 4). These differences were present in spite of the fact that the ethnic and gender composition of the control group (more young men and more black subjects) might have shifted the difference in favor of a greater bone mass in the control group. Trabecular thickness was not significantly different between patients and controls (167). These findings confirm the higher cancellous bone volume reported by Delling in primary hyperparathyroidism. However, in Delling's study, thicker trabeculae rather than increased trabecular number was the underlying structural component of the increase in bone volume (167, 176). In our study, cancellous bone volume declined with age in both normal and hyperparathyroid subjects. However, while trabecular number decreased and trabecular separation increased with age in the normal subjects, these indices showed no relationship in the hyperparathyroid subjects. Trabecular thickness, in contrast, declined with age in primary hyperparathyroidism but remained unchanged in the normal subjects. The pattern of bone loss that we observed in the normal subjects is consistent with that previously reported (198, 199). It suggests that age-related bone loss, particularly in women, occurs by the osteoclastic removal of entire trabecular plates with those remaining being more widely separated, but generally of normal thickness (198, 199, 203, 204). The pattern of bone loss observed in the hyperparathyroid subjects contrasts with that occurring in normal women after menopause and is more closely related to that described in men (198, 199, 203, 205, 206). We therefore proposed the hypothesis that the effect of increased endogenous PTH was to maintain cancellous bone volume by a mechanism that allows the relative preservation of trabecular architecture. Thus, trabecular plates, although becoming thinner with age, do not perforate and remain connected (167).

That the microarchitecture of cancellous bone is maintained was further demonstrated by comparing normal and hyperparathyroid postmenopausal women matched by age and by number of years postmenopause. A higher cancellous bone volume and an absence of age-related loss of bone structure and trabecular connectivity was again demonstrated in the hyperparathyroid group (207, 208). These studies confirm and extend the observations from Meunier's laboratory (209, 210) and strongly suggest that primary hyperparathyroidism may protect women against the postmenopausal deterioration of cancellous architecture and, possibly, therefore against vertebral fractures. Our findings on the bone structure in primary hyperparathyroidism have been confirmed recently in a separate detailed study of cancellous bone structure and turnover in 69 patients with primary hyperparathyroidism (184). Cancellous bone volume and indices of bone structure did not differ in hyperparathyroid and normal subjects. In spite of the increased cancellous bone remodeling, there was no evidence of accelerated bone loss and no disturbance of cancellous architecture other than a slight thinning of trabeculae, probably resulting from the increased activation frequency. Furthermore, in a postmenopausal subset, contrary to the age-related loss of bone mass and structure observed in normal women, no such loss was noted in the hyperparathyroid group.

Additional evidence for the clinical relevance of these bone biopsy findings is suggested by the following: in our study, bone mineral density (BMD) at the spine, a site rich in cancellous bone, was remarkably preserved, while a moderate deficit in bone mass was observed at the hip, a site of mixed cancellous and cortical composition, with a more substantial decrease at the radius, a site rich in cortical bone (157). These findings are consistent with the selective loss of cortical bone with relative maintenance of cancellous bone volume and structure in primary hyperparathyroidism. The trivial reduction of bone mass in the lumbar spine (~5%)

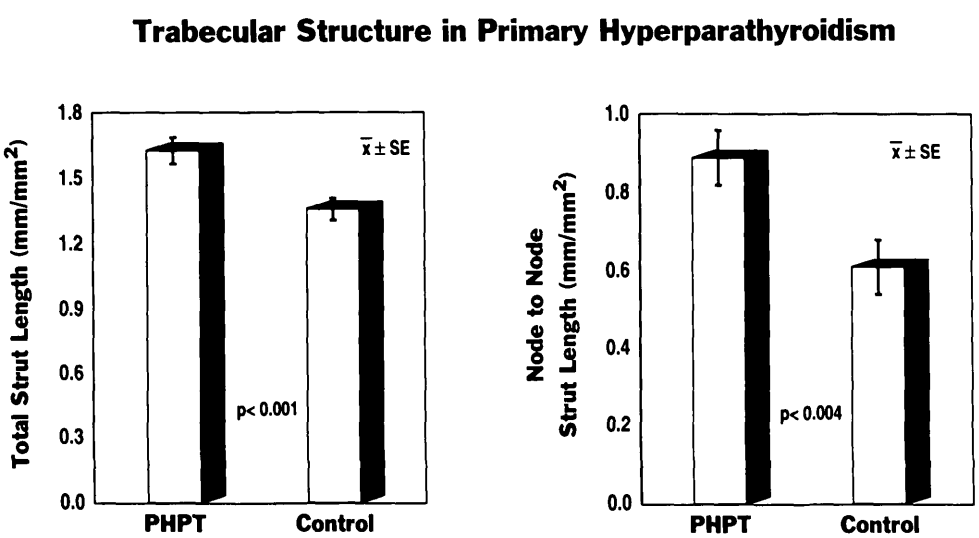

FIG. 4. Indices of trabecular structure in patients with mild primary hyperparathyroidism and controls. The higher total strut length and node to node strut length in the hyperparathyroid patients indicate higher camcellous bone volume and trabecular connectivity, respectively. [Drawn from data presented in Ref. 197a.]

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may be the net result of the relative contributions of cancellous and cortical bone when bone mass is measured noninvasively by dual photon absorptiometry (or dual energy x-ray absorptiometry). When cancellous bone was selectively measured at the iliac crest by bone biopsy, a 20% increment in this component was found. Thus, in primary hyperparathyroidism, pure cancellous bone increases while cortical bone decreases, although the possibility that cancellous bone at different skeletal sites responds differently to PTH cannot be ruled out (211). Furthermore, indices of cancellous bone mass and structure (cancellous bone volume, trabecular number, trabecular separation) at the iliac crest were correlated with BMD at the spine, with its higher cancellous component, but not at the hip and wrist at least in women with primary hyperparathyroidism (212, 213).

The cumulative data assembled in our laboratory and by others on bone mass and structure in primary hyperparathyroidism have shown that there is a maintenance or even an increase of cancellous bone volume with a conservation of cancellous microarchitecture and preserved or even increased connectivity of trabecular plates. These findings are observed not only in men and premenopausal women but also in postmenopausal women with primary hyperparathyroidism, at a stage when estrogen deficiency would normally predispose to accelerated bone loss. These data suggest that moderately increased endogenous PTH levels may protect against cancellous bone loss. There is nevertheless a concern that reductions in cortical bone mass consistently observed in this disease may offset the potential utility of PTH as a therapeutic agent in postmenopausal osteoporosis (see below). As we noted above, cortical bone loss does not occur in rats treated with intermittent PTH, and it is not clear whether this is related to the mode of administration, or species differences in response.

VI. PTH Administration in Clinical Studies

A. PTH alone

Despite the animal data (reviewed above) showing, as early as 1929, that PTE and PTH administration could have an anabolic effect on the skeleton, reports of preliminary human trials investigating PTH administration in osteoporotic patients did not appear until the mid 1970s (214, 215). This was, at least in part, because of unavailability of material. Synthetic hPTH (1–34) peptide was not available until 1974, after its synthesis according to the Niall sequence (216) by Tregear and associates (217). Bovine PTH, used therapeutically in a hypoparathyroid patient, had been shown to elicit neutralizing antibodies after short term administration (218). In preliminary data obtained from balance and kinetic studies examining the effects of short-term sc hPTH(1–34) administration, high doses (750 U daily) stimulated resorption in excess of formation, whereas lower doses (450 U) caused an overall improvement in calcium balance, with increases in net intestinal calcium and phosphate absorption, and small increases in bone accretion (215, 219). These early data were therefore comparable to those obtained in both the rat and dog experiments mentioned above.

Combined results of a small multicenter trial were published by Reeve *et al.* (96, 220, 221). A total of 21 osteoporotic patients (16 women and five men) with either vertebral or limb fractures were enrolled in seven centers. The patients and the therapies were somewhat heterogeneous, with one subject receiving pharmacological doses of vitamin D_3 for 2 yr before the study, three taking estrogen before and throughout the trial, and one man receiving testosterone throughout the study. Data were analyzed with each patient serving as his or her own control. Patients had baseline investigations including a radiocalcium kinetic study to allow determination of new bone formation and calculation of bone resorption rate, cortical densitometry of the radius or femur (by single photon absorptiometry), as well as a transiliac crest bone biopsy, with repeat studies after the treatment period.

Patients were treated with once daily sc injections of synthetic hPTH(1–34) in a dose approximately equivalent to 500 U bovine PTH(1–84) for periods of 6–24 months. No serious side effects of the treatment were noted, with only transient hypercalcemia in four cases. Overall, not surprisingly considering the heterogeneity of the group, there was a large interindividual variation in responsiveness to the treatment protocol. Mean serum total alkaline phosphatase increased 15% above baseline in association with increases in urinary hydroxyproline, calcium, and phosphate excretion. Striking increases in new bone accretion measured using a ^{47}Ca-kinetic method were noted (mean increment 144%) with lesser increments in resorption rate (220). Cortical bone density in the group as a whole did not change; however, in nine of 10 paired biopsies on which quantitative analysis could be performed, increases in cancellous bone volume (Cn-BV/TV) were seen (mean increase 92%), as well as increments in osteoid covered cancellous surfaces (mean increase 50%). The one patient who showed no increment in Cn-BV/TV was subsequently found to have developed circulating antibodies to the hPTH peptide. A close correlation was found between the ^{47}Ca accretion rate and Cn-BV/TV at the end of the treatment period.

In contrast, net intestinal calcium absorption did not increase consistently and, more importantly, overall calcium balance did not change positively. The former finding might have been due to a lack of increase in $1,25(OH)_2D$, assuming that any increment in calcium absorption would be the result of improved active transport requiring increased $1,25(OH)_2D$. This thesis was consistent with studies of the hypocalcemic induction of endogenous secondary hyperparathyroidism in which sustained increases in PTH (12 h) appeared to be necessary to stimulate increases in $1,25(OH)_2D$ (222). In two normal subjects in which the kinetics of hPTH(1–34) were studied after injection, rapid clearance from the circulation was observed with return to baseline levels after 6 h (by RIA) and after 40 min using a cytochemical assay (223). Thus, in order to augment intestinal calcium absorption, subsequent studies were performed with the addition of small daily doses of $1,25(OH)_2D$. The latter finding, *i.e.* the lack of positive calcium balance in association with promi-

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nent increments in Cn-BV/TV, led to the thesis that exogenous PTH administration in this regimen increased cancellous bone mass at the expense of cortical bone loss, a result shown in a subgroup of patients in whom bone mass had been measured in the femur (98). In order to guard against this potential problem, investigators sought to use hPTH in combination with an antiresorptive agent. In fact, the two patients who had the best response to PTH were those also on estrogen therapy (224).

B. Combination of hPTH with 1,25(OH)$_2$D

To address concern regarding lack of consistent improvement in calcium absorption, Slovik and co-workers (225, 226) added daily oral 1,25(OH)$_2$D (0.25 μg-0.5 μg) to a regimen of daily sc hPTH(1–34) (400–500 U) in eight men with idiopathic osteoporosis (mean age 50 yr) and six women with postmenopausal osteoporosis for 8–24 months. An additional 10 women received alternating 6-week cycles of hPTH(1–34) and 1,25(OH)$_2$D. There were no differences in the results between this latter group and those who received both hormones daily. A similar group of osteoporotic women treated with calcium or 1,25(OH)$_2$D alone served as controls. In four of four male subjects in whom vertebral density measurements were available, trabecular bone density (by single energy computed x-ray tomography) increased on average 98% after 12 months of therapy (Fig. 5). Increments in vertebral trabecular bone density also occurred in the women, but were not as high as those observed in the men (226). Radius density (single photon absorptiometry) did not change. In contrast, in those women treated with calcium or 1,25(OH)$_2$D alone, cortical bone density in the radius decreased. Calcium and phosphorus balance improved in four of four male patients associated with increments in dietary calcium and phosphorus absorption. The conclusion from these studies was that the combination of hPTH(1–34) and 1,25(OH)$_2$D had greater anabolic effects on bone than either agent alone in osteoporotic men and women. Given that cortical bone density did not decrease and that calcium balance increased, the authors concluded that this regimen resulted in a true increase in total bone mass without sacrificing cortical bone.

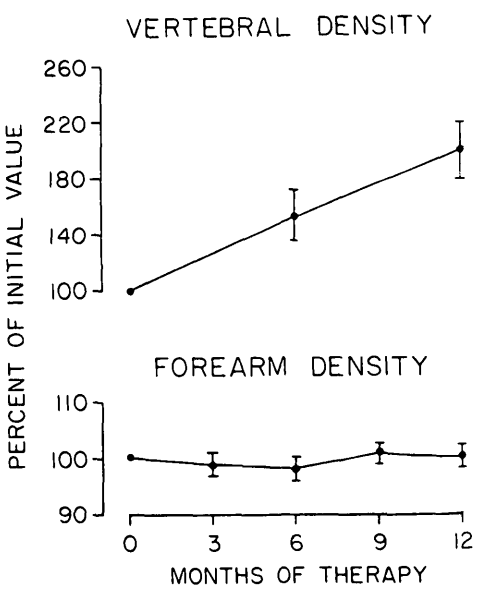

FIG. 5. Changes in trabecular bone density of lumbar vertebrae (*top*) and of cortical bone density of the radius (*bottom*) in osteoporotic men treated with daily sc injection of 400–500 U hTPH(1–34) plus daily ingestion of 15–30 mmol calcium and 0.25–5.0 μg 1,25(OH)$_2$D$_3$. [Reproduced with permission from D. M. Slovik *et al.*: *J Bone Miner Res* 1:377–381, 1986 (225).]

More recently, Neer *et al.* (227) published a controlled trial of 30 osteoporotic women, half of whom received 400–500 U hPTH (1–34) plus 0.25 μcg calcitriol/day while the control group received only calcium supplementation. Total spinal bone mineral density of PTH-treated women by dual photon absorptiometry (DPA) increased 12% over 1–2 yr with no change in the control group. In an unspecified cortical site, BMD decreased by 5.7% in the PTH-treated group *vs.* only 1.7% in the control group. Furthermore, these investigators noted that after 6–12 months, spinal bone mass stopped increasing for reasons that are unclear. There are currently no long term data on the effects of PTH treatment (>24 months). However, data from our group on the effects of PTH over an 18-month period in estrogen-treated women show a progressive increase in spinal bone mineral density without evidence of any plateau effect and without any detrimental effects on cortical bone (228).

C. Combination of hPTH with antiresorptive therapies

More recent investigations of the anabolic effect of hPTH in humans have been performed with combinations of hPTH and antiresorptive therapies. None of these studies has been well controlled or randomized, however. Modern densitometric techniques have allowed further elucidation of the effects of these regimens on different regions of the skeleton.

Hesch *et al.* (229) investigated eight osteoporotic patients treated with hPTH(1–38) instead of hPTH(1–34) as early clinical observations had suggested that the former peptide had somewhat greater potency (230). The protocol was designed to take advantage of PTH as both an activator of skeletal remodeling (see section on ADFR below) as well as an anabolic agent. 720–750 U hPTH(1–38) were given for 8 weeks, with calcitonin administered (in addition to PTH) intranasally (200 U/day) during weeks 2–4 and 6–8. Calcitonin was given alone during weeks 8–10 and no treatment was given during weeks 11–14. The entire cycle was repeated four times over a 14-month period. Density of the spine (assessed by quantitative computed tomography) increased in all subjects (12–89%). Five 'control patients' (three on no therapy, one on sodium fluoride, and one on estrogen) had no increments in spinal BMD. Forearm BMD by single photon absorptiometry (SPA) did not change in treated patients or 'controls.' Some, but not all, patients lost bone after 12 months without therapy. Overall, the combination of PTH and calcitonin appeared to be a somewhat effective anabolic regimen for the skeleton, more effective than calcitonin alone according to the results culled from the literature, but not clearly more effective than hPTH alone. Since neither hPTH nor calcitonin alone was given in the same study, these direct comparisons cannot be made.

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Reeve *et al.* (97) studied 24 patients with osteoporosis (vertebral fractures) in an open, nonrandomized 1-yr protocol (97). Twelve subjects received hPTH(1–34), and estrogen for 8 of 12 months (in nine patients) or nandrolone (25 mg every 3 weeks (in three patients). One of these patients stopped her estrogen therapy. The other 12 subjects were treated with sodium fluoride and some of this latter group included patients previously treated with parathyroid peptide. BMD of composite bone (cancellous + cortical) by DPA increased in the fluoride group approximately 13%. In the PTH + estrogen group, spinal BMD increased similarly by both quantitative computed tomography (QCT) and DPA with no change in radius BMD (by QCT) implying that the BMD increment was most likely in cancellous bone (estimated increment 50%), and that there was no cortical loss. A lack of cortical bone loss was also supported by radial bone density which was unchanged in both the PTH and the fluoride groups. This finding was similar to the lack of radial density changes found in the initial multicenter trial when hPTH(1–34) was given alone (without estrogen) but contrasted with findings in the femoral shaft where bone loss was found in a subset of patients (98).

Further investigations were performed on the patients above in the hPTH + estrogen group (231–233). Radioisotopic measures of bone formation showed marked increments (231). Calcium balance improved in most patients, with the mean increment approaching statistical significance. One patient showed a marginal increase in serum binding to hPTH(1–34) of questionable clinical significance. The combination of hPTH(1–34) and estrogen increased the width of new cancellous bone packets as well as the width of trabecular plates and there was a 20% reduction (nonsignificant) in trabecular separation (232) (Fig. 6).

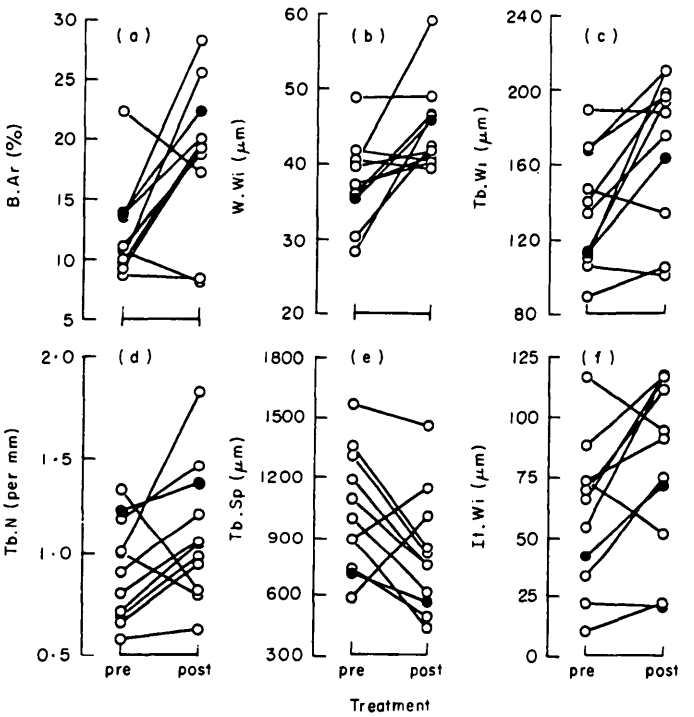

FIG. 6. Changes in trabecular bone structure in iliac crest biopsies of 11 osteoporotic women treated for 12–14 months with a combination of 460–740 "house" units of hPTH(1–34) and hormone replacement therapy (conjugated equine estrogens + norgestrel) for 12–14 months. The patient represented by *black circles* discontinued hormone replacement therapy. B.Ar, Cancellous bone area; W.Wi, wall width of trabecular bone packets; Tb.Wi, trabecular width; Tb.N, trabecular number; Tb.Sp, trabecular separation; It.Wi, interstitial width. The increments in B.Ar, W.Wi, and Tb.Wi were statistically significant ($P < 0.05$–0.01) for the group. [Reproduced with permission from J. N. Bradbeer *et al.*: *Clin Endocrinol (Oxf)* 37:282–289, 1992 (232). ©Blackwell Scientific Publications.]

D. Cyclical (ADFR) therapy

Another potential use of hPTH that must be distinguished from its daily administration to achieve a direct anabolic effect on the skeleton is its use as a skeletal activator in so-called ADFR protocols (A: activate remodeling, D: depress resorption, F: free formation, R: repeat). While both the daily use as well as the sequential use probably activate bone turnover, in the former application the peptide is administered over a long period with the hope of stimulating formation more than resorption. In the latter application, PTH is administered only for a short period to activate a cohort of remodeling units which then allows resorption to be selectively targeted by inhibiting agents. There is evidence to suggest that even intermittent injections of PTH stimulate markers of bone resorption (228, 234). Based on animal data and the observations in primary hyperparathyroidism both discussed above, however, we hypothesize that the resorption cavities formed under the influence of PTH may be shallower and not of the perforative type. Since PTH is not intended to act as an anabolic agent in the ADFR concept, such use is not really within the scope of this review. For completeness, however, the few studies that have used hPTH in ADFR protocols will be discussed.

Twelve osteoporotic patients were treated with hPTH(1–34), 1500 U/day (on the basis of studies in the greyhound dog), for 1 week followed by 3 weeks treatment with 0.25 μg $1,25(OH)_2D$ (235, 236). This cycle was then repeated 15 times. One patient developed a high titer of binding antibodies to the peptide and was thus excluded from the analysis. Mean calcium balance increased, and although individual patients had increments in trabecular bone density, there was no mean change in the group as a whole. Moreover, paired iliac bone biopsies showed no increases in Cn-BV/TV, and osteoid surface and volume and resorption surfaces all decreased. This regimen was clearly a failure, therefore, with the cyclical stimulus resulting in a paradoxical decrease rather than an increment in skeletal turnover. Possible explanations included tachyphylaxis to the activation stimulus or provocation of excess endogenous calcitonin.

Hesch *et al.* studied a heterogeneous osteoporotic group (six women, two men), three having had no previous treatment, and five having previously been treated with sodium fluoride, calcitonin, vitamin D, methenolone (an anabolic steroid), or combinations of these agents before enrollment (237). Patients received 400 U PTH(1–38) for 2 weeks. PTH(1–38) was again chosen because of its presumed greater potency (231). During the second week Etidronate (5 mg/kg)

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was also supplied for a total of 2 weeks. There was a subsequent drug-free period of 12 weeks, with the cycle repeated six times. Effects on calcium balance were highly variable, while biochemical indices reflected stimulation of bone turnover. Histological evaluation before and after treatment showed no increase in bone mass.

Another study in 20 osteoporotic patients (17 women, three men) utilized 400 IU hPTH(1–38) daily for 2 weeks (after a 24-h infusion of hPTH) as a skeletal activator followed by 8 weeks of sc salmon calcitonin (20 U/day for 1 week, 50 U/day for 1 week, and 100 U/day for 6 weeks) in half of the patients and no further treatment in the other half (238). Both groups were then given no medication for 30 days (free period) and the cycle was repeated once. $1,25(OH)_2D$ increased above baseline during the 14-day PTH treatment period. This increase was correlated with an increment in calcium absorption (by ^{45}Ca absorption test) from 24% to 30%. Serum alkaline phosphatase and osteocalcin were elevated by the end of PTH administration and remained so after an observation period of 100 days (one cycle). Smaller increases in alkaline phosphatase were seen in those patients given calcitonin. Urinary hydroxyproline was elevated only after the hPTH infusion but not during subsequent sc hPTH administration. Since biochemical changes during the calcitonin phase were so insignificant, it was felt that the PTH activation stimulus was given in too low a dose or too short a duration. The results suggested that PTH might have only modestly stimulated skeletal remodeling with persistent and more significant increases in bone formation over resorption. Bone density findings on these patients were reported in a later paper (239). After 200 days of therapy, in the PTH-treated group, vertebral bone mineral content (by DPA) increased by 13% whereas forearm BMC (by SPA) decreased by 11%. In the PTH + CT group there were no changes in bone density at either site. Histomorphometric analysis of paired biopsies performed before treatment and after two cycles revealed a significant increase in remodeling activation frequency, but no changes in Cn-BV/TV or in the thickness of trabecular bone packets.

VII. Conclusion

In conclusion, the results from human clinical trials on hPTH administration summarized in Table 2 are very prelim-

TABLE 2. Clinical studies employing PTH in the treatment of osteoporosis

Patients sex/ age range	Duration of treatment	PTH fragment dose/regimen	Coadministered agents	Results	Ref.
16♀ 49–78 5♂ 52–61	6–24 mo	Variable but mostly 400–500 U hPTH(1-34)	None	No change in calcium balance 70% increase of cancellous bone volume in iliac crest, bone loss in the distal femur	(96) (214) (215) (220) (221) (98)
8♂ 37–62	1 yr	hPTH(1-34) 400–500 U/day	0.25 μg/day $1,25(OH)_2D_3$ 600–1200 mg/day calcium	98% increase in vertebral trabecular BMD (QCT). No change in radius mid shaft BMD (SPA). Calcium retention improved by 120 mg/day	(225)
15♀	1–2 yr	hPTH(1-34) 400–500 U/day	0.25 μg/day $1,25(OH)_2D_3$ 1000–2000 mg/day calcium	32% ↑ in vertebral trabecular BMD (QCT) 12% ↑ in spinal BMD (DPA) 5.7% ↓ in cortical bone	
15♀	1–2 yr	Control	1000–2000 mg/day calcium	1.7% ↓ in cortical bone No changes in vertebral BMD	(227)
11♀/1♂ Female Mean 64 Male 67	1 yr	hPTH(1-34) 500 U/day	9F given estrogen 5–12 months 2 F/1 M given nandrolone 25 mg/3 wk	Calcium retention improved by 64 mg/day Trabecular vertebral bone density ↑ (QCT) by 50% Cancellous bone volume in iliac crest ↑ by 42%	(232) (97) (231) (233)
6♂/2♀ 33–67	14 mo	hPTH(1-38) 720–750 U/day 8 wk/13 wk cycle total of four cycles	6/13 wk intermittent nasal calcitonin 200 U/day	Vertebral BMD (QCT) ↑ by 12–89% (mean ~15%) No change in forearm BMD (SPA)	(229)

M, Male; F, female; mo, month.

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inary but do offer some hope that this agent alone or in some combination with other agents may yield a meaningful anabolic effect on the skeleton with the potential to diminish fracture frequency in patients suffering from osteoporosis. Many questions remain after the investigations reviewed above. These include: 1) To accomplish an anabolic effect of hPTH, is the addition of $1,25(OH)_2D$ required? Does PTH administered as a once daily sc injection increase $1,25(OH)_2D$ by itself and is this associated with increments in gastrointestinal calcium absorption? 2) Will hPTH administered in a dose that will not cause hypercalcemia, alone or in combination with antiresorptive drugs, cause an increase in bone mass when studied appropriately in a randomized, controlled trial? Is the combination of hPTH with antiresorptive therapy more effective than PTH alone? 3) Can bone mass in predominantly cancellous parts of the skeleton be increased without compromising cortical bone sites? 4) What is the incidence of induction of binding antibody to the hPTH amino-terminal peptide? 5) Is there any difference in efficacy between different lengths of the peptide, specifically (1–34) *vs.* (1–38)? 6) Can PTH actually help reverse the loss of trabecular connectivity seen in postmenopausal osteoporosis in addition to increasing trabecular bone volume? 7) If bone density and connectivity improve with hPTH administration, will these effects be maintained when the treatment is discontinued? 8) Finally, if changes in bone mass and structure are positive, will this translate into a reduction in fracture frequency? Much further study is needed to answer these questions.

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References

1. **Bauer W, Aub JC, Albright F** 1929 Studies of calcium phosphorus metabolism: study of bone trabeculae as ready available reserve supply of calcium. J Exp Med 49:145–162
2. **Selye H** 1932 On the stimulation of new bone-formation with parathyroid extract, irradiated ergosterol. Endocrinology 16:547–558
3. **Riggs BL, Hodgson SF, O'Fallon WM, Chao EW, Wahner HW, Muhs JM, Cedel SL, Melton III LJ** 1990 Effect of fluoride treatment on the fracture rate in postmenopausal women with osteoporosis. N Engl J Med 322:802–809
4. **Kleerekoper M, Peterson EL, Nelson DA, Phillips E, Schork MA, Tilley BC, Parfitt AM** 1991 A randomized trial of sodium fluoride as a treatment for postmenopausal osteoporosis. Osteo Int 1:155–161
5. **Neuman WF, Neuman MF, Sammon PJ, Casarett GW** 1975 The metabolism of labeled parathyroid hormone. Calcif Tissue Res 18:263–270
6. **Rouleau MF, Warshawsky H, Goltzman D** 1986 Parathyroid hormone binding *in vivo* to renal, hepatic, skeletal tissues of the rat using a radioautographic approach. Endocrinology 118:919–931
7. **Rouleau MF, Mitchell J, Goltzman D** 1988 *In vivo* distribution of parathyroid hormone receptors in bone: evidence that a predominant osseous target cell is not the mature osteoblast. Endocrinology 123:187–191
8. **Rouleau MF, Mitchell J, Goltzman D** 1990 Characterization of the major parathyroid hormone target cell in the endosteal metaphysis of rat long bones. J Bone Miner Res 5:1043–1053
9. **Rodan GA, Martin TJ** 1981 Role of osteoblasts in hormonal control of bone resorption: a hypothesis. Calcif Tissue Int 33:349–351
10. **Chambers TJ, McSheehy PMJ, Thomson BM, Fuller K** 1985 The effect of calcium-regulating hormones, prostaglandins on bone resorption by osteoclasts disaggregated from neonatal rabbit bones. Endocrinology 116:234–239
11. **McSheehy PHJ, Chambers TJ** 1986 Osteoblastic cells mediate osteoclastic responsiveness to parathyroid hormone. Endocrinology 118:824–828
12. **Perry HM, Skogen W, Chappel JC, Wilner GD, Kahn AJ, Teitelbaum S** 1987 Conditioned medium from osteoblast-like cells mediates parathyroid hormone induced bone resorption. Calcif Tissue Int 40:298–300
13. **Perry HM, Skogen W, Chappel J, Kahn AJ** 1989 Partial characterization of a parathyroid hormone-stimulated resorption factor(s) from osteoblast-like cells. Endocrinology 125:2075–2082
14. **Morris CA, Mitnick ME, Weir EC, Horowitz M, Kreider BL** 1990 The parathyroid hormone-related protein stimulates human osteoblast-like cells to secrete a 9000-dalton bone-resorbing protein. Endocrinology 126:1783–1785
15. **Fuller K, Gallagher AC, Chambers TJ** 1991 Osteoclast resorption-stimulating activity is associated with the osteoblast surface and/or the extracellular matrix. Biochem Biophys Res Commun 181:67–73
16. **Rao LG, Murray TM, Heersche JNM** 1983 Immunohistochemical demonstration of parathyroid hormone binding to specific cell types in fixed rat bone tissue. Endocrinology 113:805–810
17. **Duong LT, Grasser W, DeHaven PA, Sato M** 1990 Parathyroid hormone receptors identified on avian and rat osteoclasts. J Bone Miner Res 5 [Suppl. 2]:s203 (Abstract)
18. **Teti LT, Rizzoli R, Zambonin ZA** 1991 Parathyroid hormone binding to cultured avian osteoclasts. Biochem Biophys Res Commun 174:1217–1222
19. **Agarwala N, Gay CV** 1992 Specific binding of parathyroid hormone to living osteoclasts. J Bone Miner Res 7:531–539
20. **Reynolds JJ** 1972 Skeletal Tissue; in Bourne GH (ed): In: The Biochemistry and Physiology of Bone. Academic Press, New York, pp 70–126
21. **Herrmann-Erlee MP, Heersche JN, Hekkelman JW, Gaillard PJ, Tregear GW, Parsons JA, Potts Jr JT** 1976 Effects of bone *in vitro* of bovine parathyroid hormone, synthetic fragments representing residues 1–34, 2–34, 3–34. Endocrinol Res Commun 3:21–35
22. **Dietrich JW, Canalis EM, Maina DM, Raisz LG** 1976 Hormonal control of bone collagen synthesis *in vitro*. Effects of parathyroid hormone, calcitonin. Endocrinology 98:943–949
23. **Hefley TJ, Krieger NS, Stern PH** 1986 Simultaneous measurement of bone resorption, collagen. Ann Biochem 153:166–171
24. **Vargas SJ, Raisz LG** 1990 Simultaneous assessment of bone resorption and formation in cultures of 22-day fetal rat parietal bones: effects of parathyroid hormone and prostaglandin E_2. Bone 11:61–65
25. **Kream BE, Rowe RW, Gworek SC, Raisz LG** 1980 Parathyroid hormone alters collagen synthesis and procollagen mRNA levels in fetal rat calvariae. Proc Natl Acad Sci USA 77:5654–5658
26. **Partridge NC, Dickson CA, Kopp K, Teitlebaum SL, Crouch EC, Kahn AJ** 1989 Parathyroid hormone inhibits collagen synthesis at both ribonucleic acid and protein levels in rat osteogenic sarcoma cells. Mol Endocrinol 3:232–239
27. **Wong GL, Luben RA, Cohn DV** 1977 1,25-Dihydroxycholecalciferol and parahormone: effects on isolated osteoclast-like and osteoblast-like cells. Science 197:663–665
28. **Majeska RJ, Rodan GA** 1981 Low concentrations of parathyroid hormone enhance growth of cloned osteoblast-like cells *in vitro*. Calcif Tissue Int 33:323–000
29. **Kream BE, Rowe D, Smith MD, Maher V, Majeska R** 1986 Hormonal regulation of collagen production in a clonal rat osteosarcoma cell line. Endocrinology 119:1922–1928
30. **Beresford JN, Gallagher JA, Poser JW, Russell RGG** 1984 Synthesis of osteocalcin by human bone cells *in vitro*. Effects of 1,25 $(OH)_2D_3$, PTH and glucocorticoids. Metab Bone Dis Rel Res 5:229–234
31. **Bringhurst FR, Potts Jr JT** 1981 Bone collagen synthesis *in vitro*: structure activity relations among parathyroid hormone fragments and analogs. Endocrinology 108:103–108

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32. **DeBartolo TF, Pegg LE, Shasserre C, Hahn TJ** 1982 Comparison of parathyroid hormone and calcium inonopore A23187: effects on bone resorption and nucleic acid synthesis in cultured fetal rat bone. Calcif Tissue Int 34:495–500
33. **Canalis E, Centrella M, Burch W, McCarthy TL** 1989 Insulin-like growth factor I mediates selective anabolic effects of parathyroid hormone in bone cultures. J Clin Invest 83:60–65
34. **MacDonald BR, Gallagher JA, Russell RGG** 1986 Parathyroid hormone stimulates the proliferation of cells derived from human bone. Endocrinology 118:2445–2449
35. **Centrella M, McCarthy TL, Canalis E** 1988 Parathyroid hormone modulates transforming growth factor β activity, binding in osteoblast-enriched cell cultures form fetal rat parietal bone. Proc Natl Acad Sci USA 85:5889–5893
36. **Partridge NC, Martin TJ**, Studies on the effects of parathyroid hormone on growth of UMR 106 osteogenic sarcoma cells. American Society for Bone and Mineral Research Annual Meeting, 1984 (abstract): A19
37. **Van der Plas A, Nijweide PJ** 1988 Cell-cell interactions in the osteogenic compartment of bone. Bone 9:107–111
38. **Howard GA, Bottemiller BL, Turner RT, Raider JI, Baylink DJ** 1981 Parathyroid hormone stimulates bone formation, resorption oxygen culture: evidence for a coupling mechanism. Proc Natl Acad Sci USA 78:3204–3208
39. **McPartlin J, Skrabanek D, Powell D** 1984 Bone alkaline phosphatase: quantitative cytochemical characterization and response to parathyrin *in vitro*. Biochem Soc Trans 123:894–898
40. **Puzas JE, Brand JS** 1985 Bone cell phosphotyrosine phosphatase: characterization and regulation by calcitropic hormones. Endocrinology 116:2463–2468
41. **Nijweide PJ, Burger EH, Feyen JHM** 1986 Cells of bone: proliferation, differentiation, and hormonal regulation. Physiol Rev 66:855–886
42. **McCarthy TL, Centrella M, Canalis E** 1989 Regulatory effects of insulin-like growth factors I and II on bone collagen synthesis in rat calvarial cultures. Endocrinology 124:301–309
43. **Linkhart TA, Mohan S** 1989 Parathyroid hormone stimulates release of insulin-like growth factor-I (IGF-I), IGF-II from neonatal mouse calvaria in organ culture. Endocrinology 125:1484–1491
44. **McCarthy TL, Centrella M, Canalis E** 1989 Parathyroid hormone enhances the transcript and polypeptide levels of insulin-like growth factor I in osteoblast enriched cultures form fetal rat bone. Endocrinology 124:1247–1253
45. **Howard GA, Spencer EM** 1991 IGF-parathyroid hormone interactions on bone *in vitro* and *in vivo*. In: Spencer E (ed) Modern Concepts of Insulin-Like Growth Factors. Elsevier, New York pp 155–168
46. **Bonewald LF, Mundy GR** 1990 Role of transforming growth factor-beta in bone remodeling. Clin Orthop 250:261–276
47. **Martin TJ, Ng KW, Suda T** 1989 Bone cell physiology. Endocrinol Metab Clin N Am 18:833–858
48. **Centrella M, McCarthy TL, Canalis E** 1991 Transforming growth-factor-beta and remodeling of bone. J Bone Joint Surg 73:1418–1428
49. **Hauschka PV, Maurakos AE, Lafrati MD, Doleman SE, Klagsbrun M** 1986 Growth factors in bone matrix. J Biol Chem 261:12665–22674
50. **Robey PG, Young MF, Flanders KC, Roche NS, Kodaiah P, Reddi AH, Termine JD, Sporn MB, Roberts AB** 1987 Osteoblasts synthesize and respond to transforming growth factor-type b (TGF-B) *in vitro*. J Cell Biol 105:457–463
51. **Seitz PK, Zhu B-T, Cooper CW** 1992 Effect of transforming growth factor b on parathyroid hormone receptor binding and cAMP formation in rat osteosarcoma cells. J Bone Miner Res 7:541–546
52. **Centrella M, Massague J, Canalis E** 1986 Human Platelet-derived transforming growth factor-β stimulates parameters of bone growth in fetal rat calvariae. Endocrinology 119:2306–2312
53. **Dijke P, Iwata K, Goddard C, Pieler C, Canalis E, McCarthy TL, Centrella M** 1990 Recombinant transforming growth factor type beta 3: biological activities and receptor-binding properties in isolated bone cells. Mol Cell Biol 10:4473–4479
54. **Noda M, Camilliere J** 1989 *In vivo* stimulation of bone formation by transforming growth factor-β. Endocrinology 124:2991–2994
55. **Marcelli C, Yates AJ, Mundy GR** 1990 *In vivo* effects of human recombinant transforming growth factor beta on bone turnover in normal mice. J Bone Miner Res 5:1087–1096
56. **Mackie EJ, Trechsel U** 1990 Stimulation of bone formation *in vivo* by transforming growth factor-β: remodeling of woven bone and lack of inhibition by indomethacin. Bone 11:295–300
57. **Oreffo ROC, Mundy GR, Seyedin S, Bonewald LF** 1989 Activation of the bone derived latent TGFB complex by isolated osteoclasts. Biochem Biophys Res Commun 158:817–823
58. **Chenu C, Pfeilschifter J, Mundy GR, Roodman GD** 1988 Transforming growth factor beta inhibits formation of osteoclast-like cells in long term human marrow cultures. Proc Natl Acad Sci USA 85:5683–5687
59. **Mohan S, Baylink DJ** 1991 The role of insulin-like growth factor II in the coupling of bone formation to resorption. In: Spencer EM (ed) Modern Concepts of Insulin-Like Growth Factors. Elsevier Science Publishing Co., Loma Linda, CA, pp 169–184
60. **Zhang L, Leeman E, Carnes DC, Graves DT** 1991 Human osteoblasts synthesize and respond to platelet-derived growth factor. Am J Physiol 261:C348–C354
61. **Canalis E, Lorenzo J, Burgess WH, Maciag T** 1987 Effects of endothelial cell growth factor on bone remodelling *in vitro*. J Clin Invest 79:52–58
62. **Canalis E, McCarthy T, Centrella M** 1988 Effects of basic fibroblast growth factor on bone formation *in vitro*. J Clin Invest 81:1572–1577
63. **Tam CS, Heersche JNM, Murray TM, Parsons JA** 1982 Parathyroid hormone stimulates the bone apposition rate independently of its resorptive action: differential effects of intermittent and continuous administration. Endocrinology 110:506–512
64. **Hock JM, Hummert JR, Boyce R, Fonseca J, Raisz LG** 1989 Resorption is not essential for the stimulation of bone growth by hPTH-(1–34) in rats *in vivo*. J Bone Miner Res 4:449–458

64a. **Fujimori A, Cheng S-L, Avioli LV, Civitelli R** 1992 Structure-function relationship of parathyroid hormone: activation of phospholipase-C, protein kinase-A and -C in osteosarcoma cells. Endocrinology 130:29–36

65. **Rosenblatt M** 1986 Peptide hormone antagonists that are effective *in vivo*. N Engl J Med 315:1004–1013
66. **Rodan SB, Rodan GA** 1986 Dexamethasone effects on beta-adrenergic receptors and adenylate cyclase regulatory proteins G and G in ROS 17/2.8 cells. Endocrinology 118:2510–2518
67. **Yamaguchi DT, Hahn TJ, Iida-Klein A, Kleeman CR, Muallem S** 1987 Parathyroid hormone-activated calcium channels in an osteoblast-like clonal osteosarcoma cell line. cAMP-dependent and cAMP-independent calcium channels. J Biol Chem 262:7711–7718
68. **Cosman F, Morrow B, Kopal M, Bilezikian JP** 1989 Stimulation of inositol phosphate formation in ROS 17/2.8 cell membranes by guanine nucleotide, calcium and parathyroid hormone. J Bone Miner Res 4:413–420
69. **Fujimori A, Cheng S-L, Avioli LV, Civitelli R** 1991 Dissociation of second messenger activation by parathyroid hormone fragments in osteosarcoma cells. Endocrinology 128:3032–3039
70. **Klein RF, Nissenson RA, Strewler GJ** 1988 Forskolin mimics the effects of calcitonin but not parathyroid hormone on bone resorption *in vitro*. Bone Miner 4:247–256
71. **Murrills RJ, Dempster DW** 1990 The effects of stimulators of intracellular cyclic AMP on rat and chick osteoclasts *in vitro*: validation of a simplified light microscope assay of bone resorption. Bone 11:333–344
72. **Murrills RJ, Stein LS, Horbert WR, Dempster DW** 1992 Effects of phorbol myristate acetate on rat and chick osteoclasts. J Bone Miner Res 7:415–423
73. **Su Y, Chakraborty M, Nathanson MH, Baron R** 1992 Differential effects of the 3′,5′-cyclic adenosine monophosphate and protein kinase C pathways on the response of isolated rat osteoclasts to calcitonin. Endocrinology 131:1497–1502
74. **Teti A, Colucci S, Grano M, Argentino L, Zallone AZ** 1992 Protein kinase C affects microfilaments, bone resorption and $[Ca3^{+}]_o$ sensing in cultured osteoclasts. Am J Physiol 263:C130-C139
75. **Civitelli R, Hruska KA, Shen V, Avioli LV** 1990 Cyclic AMP-dependent and calcium dependent signals in parathyroid hormone function. Exp Gerontol 25:223–231

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76. **Reid R, Lowe C, Cornish J, Gray DH, Skinner SJM** 1990 Adenylate cyclase blockers dissociate PTH-stimulated bone resorption from cAMP production. Am J Physiol 258:E708-E714
77. **Reid IR, Civitelli R, Avioli LV, Hruska KA** 1988 Parathyroid hormone depresses cytosolic pH and DNA synthesis in osteoblast-like cells. Am J Physiol 255:E9-E15
78. **Sugimoto T, Kano J, Fukase M, Fujita T** 1992 Second messenger signaling in the regualtion of cytosolic pHand DNA synthesis by parathyroid hormone (PTH) and PTH-related peptide in osteoblastic osteosarcoma cells: role of NA^+/H^+ exchange. J Cell Physiol 152:28–34
79. **Lowik CWGM, Van Leeuwen JPTM, van der Meer JM, Van Zeeland JK, Scheven BAA, Herrmann-Erlee MPM** 1985 A two-receptor model for the action of parathyroid hormone on osteoblast: a role for intracellular free calcium and cAMP. Cell Calcium 6:311–326
80. **Berthelson S, Pettinger WA** 1977 A functional basis for the classification of alpha-adrenergic receptors. Life Sci 21:595–606
81. **Fitzpatrick LA, Coleman DT, Bilezikian JP** 1992 The target tissue actions of parathyroid hormone. In: Coe FL, Favus MJ (eds) Disorders of Bone and Mineral Metabolism. Raven Press, New York, pp 123–148
82. **Garcia JC, McConkey CL, Martin KJ** 1989 Separate binding sites for intact PTH 1–84 and synthetic PTH 1–34 in canine kidney. Calcif Tissue Int 44:214–219
83. **Finkelman RD, Mohan S, Linkhart TA, Abraham SM, Boussy JP, Baylink DJ** 1992 PTH stimulates the proliferation of TE-85 human osteosarcoma cells by a mechanism not involving either increased cAMP or increased secretion of IGF-I, IGF-II or TGF B. Bone Miner 16:89–100
84. **Murrills RJ, Stein LS, Fey CP, Dempster DW** 1990 The effects of parathyroid hormone (PTH), PTH-related peptide on osteoclast resorption of bone slices *in vitro*: an analysis of pit size and the resorption focus. Endocrinology 127:2648–2653
85. **Fenton AJ, Kemp BE, Hammonds RG, Mitchelhill K, Moseley JM, Martin TJ, Nicholson GC** 1991 A potent inhibitor of osteoclastic bone resorption within a highly conserved pentapeptide region of parathyroid hormone-related protein, PTHrP (107–111). Endocrinology 129:3424–3426
86. **Pugsley LI, Selye H** 1933 The histological changes in the bone responsible for the action of parathyroid hormone on the calcium metabolism of the rat. J Physiol (Lond) 79:113–117
87. **Jaffe HL** 1933 Hyperparathyroidism (Recklinghausen's disease of bone). Arch Pathol 16:63–112
88. **Burrows RB** 1938 Variations produced in bones of growing rats by parathyroid extracts. Am J Anat 62:237–290
89. **Shelling DH, Asher DE, Jackson DA** 1933 Calcium, phosphorus studies: effects of variations and dosage of parahormone and of calcium and phosphorus and diet on concentrations of calcium and inorganic phosphorus in serum and on histology and chemical composition of bones of rats. Bull Johns Hopkins Hosp 53:348–389
90. **Heller M, Mclean FC, Bloom W** 1950 Cellular transformations in mammalian bones induced by parathyroid extract. Am J Anat 87:315–339
91. **Kalu DN, Pennock J, Doyle FH, Foster GV** 1970 Parathyroid hormone and experimental osteosclerosis. Lancet 27:1363–1366
92. **Walker DG** 1971 The induction of osteoporotic changes in hypophysectomized, thyroparathyroidectomized and intact rats of various ages. Endocrinology 89:1389–1406
93. **Schulz A, Remagen W, Grosse P** 1971 Effects of parathyroid extract on calcium metabolism and on bone morphology in the intact rat. Z Gesamte Exp Med 155:87–97
94. **Cramer CF, Suiker AP, Copp DH** 1961 Parathyroid influence on calcium and phosphorus adsorption by the gut. In: Greep RO, Talmage RV (eds) The Parathyroids. Charles C Thomas, Springfield, IL, pp 158–166
95. **Parsons JA** 1976 Parathyroid physiology and the skeleton. In: Bourne GH (ed) The Biochemistry and Physiology of Bone. Academic Press, London, pp 159–226
96. **Reeve J, Meunier PJ, Parsons JA, Bernat M, Bijvoet OLM, Courpron P, Edouard C, Klenerman L, Neer RM, Renier JC, Slovik D, Vismans EJFE, Potts JT** 1980 Anabolic effect of human PTH on trabecular bone in involutional osteoporosis: a multicenter trial. Br Med J 280:1340–1344
97. **Reeve J, Davies UM, Hesp R, McNally E, Katz D** 1990 Treatment of osteoporosis with human parathyroid peptide and observations on effect of sodium fluoride. Br Med J 301:314–318
98. **Hesp R, Hulme P, Williams D, Reeve J** 1981 The relationship between changes in femoral bone density and calcium balance in patients with involutional osteoporosis treated with human PTH fragment 1–34. Metab Bone Dis Rel Res 2:331–334
99. **Podbesek R, Edouard C, Meunier PJ, Parsons JA, Reeve J, Stevenson RW, Zanelli JM** 1983 Effects of two treatment regimes with synthetic human parathyroid hormone fragment on bone formation and the tissue balance of trabecular bone in greyhounds. Endocrinology 112:1000–1006
100. **Podesek R, Stevenson R, Zanelli GD, Edovard C, Meunier PJ, Reeve J, Parsons JA** 1980 Treatment with human parathyroid hormone fragment (hPTH 1–34) stimulates bone formation and intestinal calcium absorption in the greyhound: comparison with data from the osteoporosis trial. In: Hormonal Control of Calcium Metabolism. Excerpta Medica, Amsterdam, pp 118–123
101. **Hefti E, Trechsel U, Bonjour JP, Fleisch H, Schenk R** 1982 Increase of whole body calcium and skeletal mass in normal and osteoporotic adult rats treated with PTH. Clin Sci 62:389–396
102. **Gunness-Hey M, Hock JM** 1984 Increased trabecular bone mass in rats treated with synthetic human PTH. Metab Bone Dis Rel Res 5:177–181
103. **Hori M, Uzawa T, Morita K, Noda T, Takahashi H, Inoue J** 1988 Effect of human parathyroid hormone (PTH(1–34)) on experimental osteopenia of rats induced by ovariectomy. Bone Miner 3:193–199
104. **Hock JM, Gera I, Fonseca J, Raisz LG** 1988 Human PTH (1–34) increases bone in ovariectomized and orchidectomized rats. Endocrinology 122:2899–2904
105. **Hock JM, Hummert J, Boyce R, Fonseca J, Raisz LG** 1989 Resorption is not essential for the stimulation of bone growth by human PTH 1–34 in rats *in vivo*. J Bone Miner Res 4:449–458
106. **Gunness-Hey M, Gera I, Fonseca J, Raisz LG, Hock JM** 1988 1,25 Dihydroxyvitamin D3 alone or in combination with parathyroid hormone does not increase bone mass in young rats. Calcif Tissue Int 43:284–288
107. **Hock JM, Fonseca J** 1990 Anabolic effect of human synthetic PTH depends on growth hormone. Endocrinology 127:1804–1810
108. **Shen V, Dempster DW, Birchman R, Xu R, Lindsay R** 1993 Loss of cancellous bone mass and connectivity in ovariectiomized rats can be restored by combined treatment with parathyroid hormone and estradiol. J Clin Invest, 91:2479–2487
109. **Tada K, Yamamuro T, Okumura H, Kasai R, Takahashi H** 1990 Restoration of axial and appendicular bone volumes by h-PTH (1-34) in parathyroidectomized and osteopenic rats. Bone 11:163–169
110. **Mellish RWE, Shen V, Birchman R, Meng XW, Horbert W, Dempster D, Lindsay R** 1990 Quantitative analysis of trabecular bone in ovariectomized rats after intermittent administration of low dose human parathyroid hormone fragment (1–34) and 17β-estradiol. In: Christiansen C, Overgaard K (eds) Osteoporosis 1990. Handestrykkeriet Aalborg APS, Aalborg, Denmark, pp 1335–1337
111. **Gunness-Hey M, Hock JM** 1989 Loss of the anabolic effect of PTH on bone after discontinuation of hormone in rats. Bone 10:447–452
112. **Liu CC, Kalu DN** 1990 Human parathyroid hormone (1–34) prevents bone loss, augments bone formation in sexually mature ovariectomized rats. J Bone Miner Res 5:973–982
113. **Liu CC, Kalu DN, Salerno E, Echon R, Hollis BW, Ray M** 1991 Preexisting bone loss associated with ovariectomy in rats is reversed by PTH. J Bone Miner Res 6:1071–1080
114. **Mosekilde L, Sogaard CH, Danielson CC, Torring O, Nilsson MHL** 1991 The anabolic effects of human PTH on rat vertebral body mass are also reflected in the quality of bone, assessed by biomechanical testing: a comparison study between hPTH (1–34) and hPTH (1–84). Endocrinology 129:421–428
115. **Wronski TJ, Cintron M, Dann LM** 1988 Temporal relationship between bone loss and increased bone turnover in ovariectomized rats. Calcif Tissue Int 43:179–183
116. **Ibbotson KJ, Orcutt CM, D'Souza SM, Paddock CL, Arthur JA, Jankowsky ML, Boyce RW** 1992 Contrasting effects of PTH and

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IGF-1 in an aged OVX rat model of postmenopausal osteoporosis. J Bone Miner Res 7:425–432
117. **Tuukkanen J, Jalovaara P, Vaananen K** 1990 Calcitonin treatment of immobilization osteoporosis rats. Acta Physiol Scand 141:119–124
118. **Kalu DN, Echon R, Hollis BW** 1990 Modulation of ovariectomy-related bone loss by parathyroid hormone in rats. Mech Ageing Dev 56:49–62
119. **Ejeersted C, Andreassen TT, Oxlund H, Jorgensen PH, Bak B, Nilsson M, Torring O,** PTH (1–34) and (1–84) increase the mechanical strength and thickness of cortical bone. Program of the Second Rat Skeleton Workshop, Chicago, IL, 1992 (Abstract)
120. **Kalu DN** 1991 The ovariectomized rat model of postmenopausal bone loss. Bone Miner 15:175–191
121. **Frost HM, Jee WSS** 1992 On the rat model of human osteopenias and osteoporosis. Bone Miner 18:227–236
122. **Malluche HH, Sherman D, Meyer W, Ritz E, Norman AW, Massry SG** 1982 Effects of long term infusion of physiologic doses of 1–34 PTH on bone. Am J Physiol 242:F197–F202
123. **Hock JM, Gera I** 1992 Effects of continuous and intermittent administration and inhibition of resorption on the anabolic response of bone to PTH. J Bone Miner Res 7:65–72
124. **Cheng PT, Bader SM** 1990 Effects of fluoride on rat cancellous bone. Bone Miner 11:153–161
125. **Mori S, Jee WSS, Li XJ, Chan S, Kimmel DB** 1990 Effects of prostaglandin E_2 on production of new cancellous bone in the axial skeleton of ovariectomized rats. Bone 11:103–113
126. **Ke HZ, Li XJ, Jee WSS** 1991 Partial loss of anabolic effect of prostaglandin E_2 on bone after its withdrawal in rats. Bone 12:173–183
127. **Erben RG, Weiser H, Sinowatz F, Rambeck WA, Zucker H** 1992 Vitamin D metabolites prevent vertebral osteopenia in ovariectomized rats. Calcif Tissue Int 50:228–236
128. **Jerome CP,** Comparison of effects of growth hormone (GH), growth hormone releasing factor peptide (GRF), transforming growth factor beta 2 (TGF beta), PTH and PTH+GH on bone formation and bone mass in retired breeder female rats. Program of the Second Rat Skeleton Workshop, Chicago, IL, 1992 (Abstract)
129. **Muller K, Cortesi R, Jeker H,** Effects of PTH, IGF-1 and TGF beta 2 on trabecular and cortical bone in the intact and ovx rats. Program of the Second Rat Skeleton Workshop, Chicago, IL, 1992 (Abstract)
130. **Theun HM, Roholl PJM, Spencer EM,** PTH, IGF-1 effects in an aging rat model for postmenopausal osteoporosis. Program of the Second Rat Skeleton Workshop, Chicago, IL, 1992 (Abstract)
131. **Clemmons DR** 1992 Editorial: role of insulin-like growth factor-1 in reversing catabolism. J Clin Endocrinol Metab 75:1183–1185
132. **McGuire JL, Marks SCJ** 1974 The effects of PTH on bone cell structure and function. Clin Orthop 100:392–405
133. **Gera I, Hock JM, Gunness-Hey M, Fonseca J, Raisz LG** 1987 Indomethacin does not inhibit the anabolic effect of PTH on the long bones of rats. Calcif Tissue Int 40:206–211
134. **Pocock NA, Eisman JA, Gwinn T, Sambrook P, Kelly P, Freund M, Yeates M** 1989 Muscle strength, physical fitness but not age predict femoral neck bone mass. J Bone Miner Res 4:441–448
135. **Norman AW** 1985 The vitamin D endocrine system. Physiologist 28:219–232
136. **Lawson DEM, Muir E** 1991 Molecular biology and vitamin D function. Proc Nutr Soc 50:131–137
137. **Lindgren U, DeLuca HF** 1982 Role of parathyroid hormone and 1,25-dihydroxyvitamin D_3 in the development of osteopenia in oophorectomized rats. Calcif Tissue Int 34:510–514
138. **Tam CS, Jones G, Heersche JNM** 1981 The effect of vitamin D restriction and repletion on bone apposition in the rat and its dependence on PTH. Endocrinology 109:1448–1453
138a. **Dempster DW, Lindsay R** 1993 Pathogenesis of osteoporosis. Lancet 341:797–801
138b. 1993 Consensus Development Conference Statement from the Fourth International Symposium on Osteoporosis. Am J Med, in press
139. **von Recklinghausen F** 1891 Die fibrose oder deformirende Ostitis die Osteomalacie und die carcinose in gengensitigen. In: Greiner G (ed) Beziehungen. Festschrift Rudolf Virchow, Berlin
140. **Mandl R** 1925 Therapeutisher verusch bei ostitis fibrosa generalizata mittels extirpation eines epithelkorperchen tumors. Wien Klin Wochenschr 38:1383
141. **Albright F, Aub JC, Bauer W** 1934 Hyperparathyroidism: a common and polymorphic condition as illustrated by seventeen proven cases from one clinic. JAMA 102:1276
142. **Albright F, Reifenstein EC** 1948 The Parathyroid Glands and Metabolic Bone Disease. Williams & Wilkins, Baltimore, MD
143. **Murray TM** 1989 Parathyroid hormone and hyperparathyroidism. In: Heersche JNM (ed) Metabolic Bone Disease: Cellular and Tissue Mechanisms. CRC Press, Boca Raton, FL, pp 105–133
144. **Uehlinger E** 1940 Osteofibrosis deformans juvenilis. Virchows Arch [A] 306:255
145. **Jaffe HL** 1968 Tumors and Tumorous Conditions of the Bone and Joints. Lea & Febiger, Malvern, PA
146. **Schajowicz F** 1981 Tumors and Tumor-like Lesions of Bones and Joints. Springer Verlag, Berlin
147. **Parfitt AM** 1976 The actions of parathyroid hormone on bone. Metabolism 25:1033–1069
148. **Parfitt AM, Oliver I, Villanueva ADR** 1979 Bone histology in metabolic bone disease. Orthop Clin North Am 10:329–345
149. **Mallette LE, Bilezikian JP, Heath DA, Aurbach GD** 1974 Primary hyperparathyroidism: clinical and biochemical features. Medicine 53:127–146
150. **Keating RF** 1961 Diagnosis of primary hyperparathyroidism. JAMA 178:547–555
151. **Boonstra CE, Jackson CE** 1965 Hyperparathyroidism detected by routine serum calcium analysis. Ann Intern Med 63:468–474
152. **Purnell DC, Smith LH, Scholz DA** 1971 Primary hyperparathyroidism: a prospective clinical study. Am J Med 50:670–678
153. **Lafferty FW** 1981 Primary hyperparathyroidism: changing clinical spectrum, prevalence of hypertension and discriminant analysis of laboratory tests. Arch Intern Med 141:1761–1766
154. **Mundy GR, Cove DH, Fisken R** 1980 Primary hyperparathyroidism: changes in the pattern of clinical presentation. Lancet 1:1318–1320
155. **Heath H, Hodgson SF, Kennedy MA** 1980 Primary hyperparathyroidism incidence, morbidity, and potential economic impact in a community. N Engl J Med 302:189–193
156. **Rao SD** 1985 Primary hyperparathyroidism: changing patterns in presentation and treatment decisions in the eighties. Henry Ford Hosp Med J 33:194–197
157. **Dempster DW, Feldman F, Seldin D, Jacobs TP, Sirise Cafferty M, Parisien MV, Lindsay R, Clemens TL, Bilezikian JP** 1989 Skeletal disease in primary hyperparathyroidism. J Bone Miner Res 4:283–291
158. **Dauphine RT, Riggs BL, Scholz DA** 1975 Back pain and vertebral crush fractures: an unemphasized mode of presentation for primary hyperparathyroidism. Ann Intern Med 83:365–367
159. **Kochersberger G, Buckley NJ, Leight GS, Martinez S, Studenski S, Vogler J, Lyles KW** 1987 What is the clinical significance of bone loss in primary hyperparathyroidism. Arch Intern Med 147:1951–1953
160. **Peacock M, Horsman A, Aaron JE, Marshall DH, Selby PL, Simpson M** 1984 The role of parathyroid hormone in bone loss. In: Christiansen C, Arnaud CD, Nordin BEC, Parfitt AM, Peck WA, Riggs BL (eds): Osteoporosis. Proceedings of the 1984 Copenhagen International Symposium on Osteoporosis. Aalborg Stiftsbogtrykkeri, Copenhagen, pp 463–467
161. **Martin P, Bergmann P, Gillet C, Fuss M, Kinnaert P, Corvilain J, van Geertrugden J** 1986 Partially reversible osteopenia after surgery for primary hyperparathyroidism. Arch Intern Med 146:689–691
162. **Richardson ML, Pozzi-Mucelli RS, Kanter AS, Kolb FO, Ettinger B, Genant HK** 1986 Bone mineral changes in primary hyperparathyroidism. Skeletal Radiol 15:85–95
163. **Kleerekoper M, Villaneuva AR, Mathews CHE** 1983 PTH mediated bone loss in primary, secondary hyperparathyroidism. In: Frame B, Potts JT (eds) Clinical Disorders of Bone and Mineral Metabolism. Excerpta Medica, Amsterdam, pp 200–203
164. **Rao DS, Wilson RJ, Kleerekoper M, Parfitt AM** 1988 Lack of biochemical progression or continuation of accelerated bone loss in mild asymptomatic primary hyperparathyroidism: evidence for biphasic disease course. J Clin Endocrinol Metab 67:1294–1298
165. **Parfitt AM** 1986 Accelerated cortical bone loss: primary and

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secondary hyperparathyroidism. In: Uhthoff H, Stahl E (eds) Current Concepts of Bone Fragility. Springer Verlag, Berlin, pp 279–285

166. **Parfitt AM** 1989 Surface specific bone remodeling in health and disease. In: Kleerekoper M (ed) Clinical Disorders of Bone and Mineral Metabolism. Mary Ann Liebert, Inc., New York, pp 7–14
167. **Parisien M, Silverberg SJ, Shane E, de la Cruz L, Lindsay R, Bilezikian JP, Dempster DW** 1990 The histomorphometry of bone in primary hyperparathyroidism: preservation of cancellous bone structure. J Clin Endocrinol Metab 70:930–938
168. **Seeman E, Wahner HW, Offord KP, Kumar R, Johnson WJ, Riggs BL** 1982 Differential effects of endocrine dysfunction on the axial and the appendicular skeleton. J Clin Invest 69:1302–1309
169. **Wilson RJ, Rao DS, Ellis B, Kleerekoper M, Parfitt AM** 1988 Mild asymptomatic primary hyperparathyroidism is not a risk factor for vertebral fractures. Ann Intern Med 109:959–962
170. **Lafferty FW, Hubay CA** 1989 Primary hyperparathyroidism: a review of the long term surgical and non-surgical morbidities as a basis for a ration approach to treatment. Arch Intern Med 149:789–796
171. **Melton III LJ, Atkinson EJ, O'Fallon WM, Heath HI** 1992 Risk of age-related fractures in patients with primary hyperparathyroidism. Arch Intern Med 152:2269–2273
172. **Mosekilde L, Melsen F** 1978 A tetracycline-base histomorphometric evaluation of bone resorption and bone turnover in hyperthyroidism and hyperparathyroidism. Acta Med Scand 204:97–102
173. **Melsen F, Mosekilde L** 1981 The role of bone biopsy in the diagnosis of metabolic bone disease. Orthop Clin North Am 12:571–602
174. **Eriksen EF** 1986 Normal, pathological remodeling of human trabecular bone: three dimensional reconstruction of the remodeling sequence in normals and in metabolic bone disease. Endocr Rev 7:379–408
175. **Delmas PD, Meunier PJ, Faysse E, Saubier EC** 1986 Bone histomorphometry and serum bone gla-protein in the diagnosis of primary hyperparathyroidism. World J Surg 10:572–578
176. **Delling G** 1987 Bone morphology in primary hyperparathyroidism. A qualitative and quantitative study of 391 cases. Appl Pathol 55:147–159
177. **de Vernejoul MC, Benamount MP, Cancela L** 1988 Hyperparathyroidie primitive vue en rhumatologie: signes cliniques et relations entre les signes histologiques osseux et les parametres biologiques. Rev Rhum 55:489–494
178. **Baron R, Magee SS** 1983 Estimation of trabecular bone resorption by histomorphometry: evidence for a prolonged reversal phase with normal resorption in post menopausal osteoporosis and coupled increased resorption in primary hyperparathyroidism. In: Frame B, Potts JT (eds) Clinical Disorders of Bone and Mineral Metabolism. Excerpta Medica, Amsterdam, pp 191–195
179. **Riggs BL, Kelly PJ, Jowsey J, Keating FR** 1965 Skeletal alterations in hyperparathyroidism: determination of bone formation, resorption, and morphologic changes by microradiography. J Clin Endocrinol Metab 25:777–783
180. **Byers PD, Smith R** 1971 Quantitative histology of bone in hyperparathyroidism: its relation to clinical features, x-ray, and biochemistry. Q J Med 160:471–486
181. **Wilde CD, Jaworski ZF, Villaneuva AR, Frost HM** 1973 Quantitative histological measurements of bone turnover in primary hyperparathyroidism. Calcium Tissue Res 12:137–142
182. **Meunier PJ, Vignon G, Bernard J** 1973 Clinical aspects of metabolic bone disease. In: Frame B, Parfitt AM, Duncan H (eds) Quantitative Bone Histology as Applied to the Diagnosis of Hyperparathyroid States. Excerpta Medica, Amsterdam, pp 215–221
183. **Eriksen EF, Mosekilde L, Melsen F** 1973 Trabecular bone remodeling and balance in primary hyperparathyroidism. Bone 7:213–221
184. **Christiansen P, Steiniche T, Vesterby A, Mosekilde L, Hessov I, Melsen F** 1992 Primary hyperparathyroidism: iliac crest trabecular bone volume, structure, remodeling, and balance evaluated by histomorphometric methods. Bone 13:41–49
185. **Doyle FH** 1966 Some quantitative radiological observations in primary and secondary hyperparathyroidism. Br J Radiol 39:161–167
186. **Ellis K, Hostim RJ** 1960 The skull in hyperparathyroid bone disease. Am J Roentgenol Radium Ther Nucl Med 83:732–740
187. **Jaffe HL** 1933 Hyperparathyroidism (Recklinghausen's disease of bone). Arch Pathol 16:63–112
188. **Aitken RE, Kerr JL, Lloyd HM** 1964 Primary hyperparathyroidism with osteosclerosis and calcification in articular cartilage. Am J Med 37:813–820
189. **Eugenidis N, Olah AJ, Haas HG** 1972 Osteosclerosis in hyperparathyroidism. Radiology 105:265–275
190. **Genant HK, Baron JM, Straus FH, Paloyan E, Jowsey J** 1975 Osteosclerosis in primary hyperparathyroidism. Am J Med 59:104–113
191. **van Holsbeeck M, Roex L, Favril A, Burssens D, Baert AL** 1987 Osteosclerosis in primary hyperparathyroidism. Fortschr Rontgenstr 147:690–691
192. **Dresser R** 1933 Osteitis fibrosa cystica associated with parathyroid overactivity. Am J Roentgenol Radium Ther Nucl Med 30:596–603
193. **Albright F, Baird PC, Cope D, Bloomberg E** 1934 Studies on the physiology of the parathyroid glands. IV. Renal complication of hyperparathyroidism. Am J Med Sci 187:49–65
194. **Aub JC**, Lack of effect of parathyroid gland on growth of bone: case presentation. In: Proceedings of the 26th Annual Meeting of the Endocrine Society, Atlantic City, NJ, 1942, p 1024
195. **Shallow TA, Fry KE** 1948 Parathyroid adenoma. Occurrence in father and daughter. Surgery 24:1020–1025
196. **Adam A, Ritchie D** 1954 Hyperparathyroidism with increased bone density in the areas of growth. J Bone Joint Surg 36B:257–260
197. **Lloyd HM, Aitken RE, Ferrier TM** 1965 Primary hyperparathyroidism resembling rickets of late onset. Br Med J 2:853–856

197a. **Parisien M, Mellish RWE, Silverberg SJ, Shane E, Lindsay R, Bilezikian JP, Dempster DW** 1992 Maintenance of cancellous bone connectivity in primary hyperparathyroidism: trabecular strut analysis. J Bone Miner Res 7:913–919

198. **Wakamatsu E, Sissons HA** 1969 The cancellous bone of the iliac crest. Calcif Tissue Res 4:147–161
199. **Parfitt AM, Mathews HE, Villaneuva AR, Kleerekoper M, Frame B** 1983 Relationships between surface, volume, and thickness of iliac trabecular bone in aging and in osteoporosis. Implications for the microanatomic and cellular mechanisms of bone loss. J Clin Invest 72:1396–1409
200. **Garrahan N, Mellish R, Compston J** 1986 A new method for the two-dimensional analysis of bone structure in human iliac crest biopsies. J Microsc 142:341–349
201. **Mellish RWE, Ferguson-Pell MW, Cochran GVB, Lindsay R, Dempster DW** 1991 A new method for the two-dimensional analysis of bone structure in human iliac crest biopsies. J Bone Miner Res 6:689–696
202. **Vesterby A** 1990 Star volume of marrow space and trabeculae in iliac crest. Sampling procedure and correlation to star volume of first lumbar vertebra. Bone 11:149–155
203. **Parfitt AM** 1987 Trabecular bone architecture in the pathogenesis and prevention of fracture. Am J Med 82:68–72
204. **Parisien MV, McMahon D, Pushparaj N, Dempster DW** 1988 Trabecular architecture in iliac crest bone biopsies: intra-individual variability in structural parameters and changes with age. Bone 9:289–295
205. **Kleerekoper M, Villanueva AR, Stanciu J, Rao DS, Parfitt AM** 1985 The role of three-dimensional trabecular microstructure in the pathogenesis of vertebral compression fractures. Calcif Tissue Int 37:594–597
206. **Aaron JE, Makins NB, Sagreiya K** 1987 The microanatomy of trabecular bone loss in normal aging men and women. Clin Orthop 215:260–271
207. **Parisien M, Recker R, Silverberg SJ, Shane E, Mellish RWE, Lindsay R, Bilezikian JP, Dempster DW** 1990 Cancellous bone structure in postmenopausal women with primary hyperparathyroidism. In: Christiansen C, Overgaard K (eds) Third International Symposium on Osteoporosis. Osteopress ApS, Copenhagen, pp 1139–1140
208. **Parisien M, Schnitzer M, Nieves J, Mellish RW, Silverberg SJ, Shane E, Recker R, Kimmel D, Cosman F, Bilezikian JP, Lindsay R, Dempster DW,** A comparison of bone structure and turnover in postmenopausal women with osteoporosis or primary hyper-

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parathyroidism. In: Proceedings of the Fourth International Symposium on Osteoporosis, Hong Kong, 1993, pp 162–163
209. **Courpron P, Meunier P, Bressot C, Giroux JM** 1976 Amount of bone in iliac crest biopsy: significance of the trabecular bone volume. Its values in normal and in pathological conditions. In: Meunier PJ (ed) Proceedings of Bone Histomorphometry. Lyon Second International Workshop, Lyon, France, pp 39–53
210. **Charhon SA, Edouard CM, Arlot ME, Meunier PJ** 1982 Effects of parathyroid hormone on remodeling of iliac trabecular bone packets in patients with primary hyperparathyroidism. Clin Orthop 162:255–263
211. **Hesp R, Reeve J** 1990 Letter to the editor. J Bone Miner Res 5:103
212. **Parisien M, Dempster DW, Silverberg SJ, Shane E, Seldin D, Lindsay R, Bilezikian JP** 1990 Relationship between bone mass as assessed by densitometry and by histomorphometry in primary hyperparathyroidism. J Bone Miner Res 5:[Suppl 2]:S108
213. **Parisien M, Cosman F, Silverberg SJ, Shane E, Lindsay R, Bilezikian JP, Dempster DW** 1992 Relationship between bone mass by densitometry and by histomorphometry in primary hyperparathyroidism and in osteoporosis. Bone Miner 17[Suppl 1]:214(Abstract)
214. **Reeve J, Hesp R, Williams D, Hulme P, Klenerman L, Zanelli JM, Darby AJ, Tregear GW, Parsons JA** 1976 Anabolic effect of low doses of a fragment of human parathyroid hormone on the skeleton in postmenopausal osteoporosis. Lancet 1:1035–1038
215. **Reeve J, Tregear GW, Parsons JA** 1976 Preliminary trial of low doses of human parathyroid hormone 1–34 peptide in treatment of osteoporosis. Calcif Tissue 21:469–477
216. **Niall HD, Sauer RT, Jacobs JW, Keutmann HT, Segre GV, O'Riordan JLH, Aurbach GD, Potts Jr JT** 1974 The amino acid sequence of the amino terminal 37 residues of human parathyroid hormone. Proc Natl Acad Sci USA 71:384–388
217. **Tregear GW, VanRietschoten J, Greene E, Keutmann HT, Niall HD, Reit B, Parsons JA, Potts Jr JT** 1974 Bovine parathyroid hormone: minimum chain length of synthetic peptide required for biological activity. Endocrinology 93:1349–1353
218. **Melick RA, Gill Jr JR, Berson SA, Yalow RS, Bartter FC, Potts Jr JT, Aurbach GD** 1967 Antibodies and clinical resistance to parathyroid hormone. N Engl J Med 276:144–147
219. **Slovik DM, Neer RM, Potts Jr JT** 1981 Short-term effects of synthetic human parathyroid hormone (1–34) administration on bone mineral metabolism in osteoporotic patients. J Clin Invest 68:1261–1271
220. **Reeve J, Arlot M, Bernat M, Charhon S, Edouard C, Slovik D, Vismans FJFE, Meunier PJ** 1981 Calcium-47 kinetic measurements of bone turnover compared to bone histomorphometry in osteoporosis: the influence of human parathyroid fragment (hPTH 1–34) therapy. Metab Bone Dis Rel Res 3:23–30
221. **Reeve J, Arlot M, Bernat M, Edouard C, Hesp R, Slovik D, Vismans FJFE, Meunier PJ** 1980 Treatment of osteoporosis with human parathyroid hormone fragment 1–34: a positive final tissue balance in trabecular bone. Metab Bone Dis Rel Res [Suppl] 2:355
222. **Bilezikian JP, Canfield RE, Jacobs TP, Polay JS, D'Amano P, Eisman JA, DeLuca HF** 1978 Response of 1,25-dihydroxyvitamin D_3 to hypocalcemia in human subjects. N Engl J Med 299:437–441
223. **Kent GN, Loveridge N, Reeve J, Zanelli JM** 1985 Pharmacokinetics of synthetic human parathyroid hormone 1–34 in man measured by cytochemical bioassay and radioimmunoassay. Clin Sci 68:171–177
224. **Parsons JA, Meunier PJ, Neer RM, Podbesek R, Reeve J** 1981 Effects of synthetic human parathyroid hormone fragment (hPTH 1–34) on bone mass and bone mineral metabolism. In: DeLuca HF, Frost HM, Jee JSS, Johnston Jr CC, Parfitt AM (eds) Osteoporosis. University Park Press, Baltimore, MD, pp 457–465
225. **Slovik DM, Rosenthal DI, Doppelt SH, Potts Jr JT, Daly MA, Campbell JA, Neer RM** 1986 Restoration of spinal bone in osteoporotic men by treatment with human parathyroid hormone (1–34) and 1,25-dihydroxyvitamin D. J Bone Miner Res 1:377–381
226. **Neer RM, Slovik D, Doppelt S** 1987 The use of parathyroid hormone plus 1,25-dihydroxyvitamin D to increase trabecular bone in osteoporotic men and postmenopausal women. In: Christiansen C, Johansen JS, Riis BJ (eds): Osteoporosis 1987. Osteopress, Copenhagen, pp 829–835
227. **Neer M, Slovik DM, Daly N, Potts Jr JT, Nussbaum SR** 1993 Treatment of postmenopausal osteoporosis with daily parathyroid hormone plus calcitriol. Osteopor Int [Suppl] 1:S204–205
228. **Lindsay R, Cosman F, Nieves J, Dempster DW, Shen VA** 1993 Controlled trial of the effects of 1–34hPTh in estrogen treated osteoporotic women. J Bone Miner Res (Abstract), 8[Suppl. 1]:S130
229. **Hesch R-D, Busch U, Prokop M, Delling G, Rittinghaus E-F** 1989 Increase of vertebral density by combination therapy with pulsatile 1–38hPTH and sequential addition of calcitonin nasal spray in osteoporotic patients. Calcif Tissue Int 44:176–180
230. **Hesch RD, Heck J, Auf'mkolk B** 1984 First clinical observations with hPTH (1–38), a more potent human parathyroid hormone peptide. Horm Metab Res 169:559–560
231. **Reeve J, Bradbeer JN, Arlot M, Davies UM, Green JR, Hampton L, Edouard C, Hesp R, Hulme P, Ashby JP, Zanelli JM, Meunier PJ** 1991 hPTH 1–34 treatment of osteoporosis with added hormone replacement therapy: biochemical, kinetic and histological responses. Osteopor Int 1:162–170
232. **Bradbeer JN, Arlot ME, Meunier PJ, Reeve J** 1992 Treatment of osteoporosis with parathyroid peptide (hPTH 1–34) and oestrogen: increase in volumetric density of iliac cancellous bone may depend on reduced trabecular spacing as well as increased thickness of packets of newly formed bone. Clin Endocrinol (Oxf) 37:282–289
233. **Reeve J, Arlot ME, Bradbeer JN, Hesp R, McNally E, Meunier PJ, Zanelli JM** 1993 Human parathyroid peptide treatment of vertebral osteoporosis. Osteopor Int 3[Suppl]:199–203
234. **Hodsman AB, Fraher LJ, Ostbye T, Adachie JD, Steer BM** 1993 An evaluation of several biochemical markers for bone formation and resorption in a protocol utilizing cyclical parathyroid hormone and calcitonin therapy for osteoporosis. J Clin Invest, 91:1138–1148
235. **Reeve J, Arlot M, Price TR, Edouard C, Hesp R, Hulme P, Ashby JP, Zanelli JM, Green JR, Tellez M, Katz D, Spinks J, Meunier PJ** 1987 Periodic courses of human 1–34 parathyroid peptide alternating with calcitriol paradoxically reduce bone remodeling in spinal osteoporosis. Eur J Clin Invest 17:421–428
236. **Reeve J, Podbesek RD, Price TR, Arlot M, Bartlett C, Deacon AC, Edovard C, Green JR, Hesp R, Hulme P, Katz D, Tellez M, Zanelli GD, Zanelli JM, Meunier PJ** 1984 Studies of a "short-cycle" ADFR regime using parathyroid peptide hPTH 1–34 in idiopathic osteoporosis and in a dog model. In: Christiansen C, Arnaud CD, Nordin BEC, *et al.* (eds): Osteoporosis 1984. Aalborg Stiftsbogtrykkeri, Copenhagen, pp 567–573
237. **Hesch RD, Heck J, Delling G, Keck E, Reeve J, Canzler H, Schober O, Harms H, Rittinghaus EF** 1988 Results of a stimulatory therapy of low bone metabolism in osteoporosis with (1–38)hPTH and disphosphonate EHDP. Klin Wochenschr 66:976–984
238. **Hodsman AB, Fraher LJ** 1990 Biochemical responses to sequential human parathyroid hormone (1–38) and calcitonin in osteoporotic subjects. Bone Miner 9:137–152
239. **Hodsman AB, Steer BM, Fraher LJ, Drost DJ** 1991 Bone densitometric and histomorphometric responses to sequential human parathyroid hormone (1–38) and salmon calcitonin in osteoporotic patients. Bone Miner 14:67–83

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